

Lorazepam: Drug information - UpToDate

Lorazepam: Drug information - UpToDate

Official reprint from UpToDate®

www.uptodate.com © 2026 UpToDate, Inc. and/or its affiliates. All Rights Reserved.

Lorazepam: Drug information

2026© UpToDate, Inc. and its affiliates and/or licensors. All Rights Reserved.

Contributor Disclosures

For additional information see "Lorazepam: Patient drug information" and "Lorazepam: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Risks from concomitant use with opioids (injection, oral):

Concomitant use of benzodiazepines and opioids may result in profound sedation, respiratory depression, coma, and death. Reserve concomitant prescribing of these drugs for patients for whom alternative treatment options are inadequate. Limit dosages and durations to the minimum required. Follow patients for signs and symptoms of respiratory depression and sedation.

Abuse, misuse, and addiction (injection, oral):

The use of benzodiazepines, including lorazepam, exposes users to risks of abuse, misuse, and addiction, which can lead to overdose or death. Abuse and misuse of benzodiazepines commonly involve concomitant use of other medications, alcohol, and/or illicit substances, which is associated with an increased frequency of serious adverse outcomes. Before prescribing lorazepam and throughout treatment, assess each patient's risk for abuse, misuse, and addiction.

Dependence and withdrawal reactions:

Injection: The continued use of benzodiazepines for several days to weeks may lead to clinically significant physical dependence. The risks of dependence and withdrawal increase with longer treatment duration and higher daily dose. Although lorazepam injection is indicated only for intermittent use, if used more frequently than recommended, abrupt discontinuation or rapid dosage reduction of lorazepam injection may precipitate acute withdrawal reactions, which can be life-threatening. For patients using lorazepam injection more frequently than recommended, to reduce the risk of withdrawal reactions, use a gradual taper to discontinue lorazepam injection.

Oral: The continued use of benzodiazepines, including lorazepam, may lead to clinically significant physical dependence. The risks of dependence and withdrawal increase with longer treatment duration and higher daily dose. Abrupt discontinuation or rapid dosage reduction of lorazepam after continued use may precipitate acute withdrawal reactions, which can be life-threatening. To reduce the risk of withdrawal reactions, use a gradual taper to discontinue lorazepam or reduce the dosage.

Brand Names: US

- Ativan;
- LORazepam Intensol;

- Loreev XR

Brand Names: Canada

- APO-LORazepam;
- Ativan;
- PMS-LORazepam;
- PRO-LORazepam;
- TEVA-LORazepam

Pharmacologic Category

- Antiseizure Agent, Benzodiazepine;
- Benzodiazepine

Dosing: Adult

Dosage guidance:

Safety: Reduce dose or avoid use in patients receiving opioids or with significant chronic disease (eg, respiratory compromise). Avoid use in patients with a history of substance use, misuse of medications, or depression, except for acute or emergency situations (eg, acute agitation, status epilepticus) (Ref).

Akathisia, antipsychotic induced

Akathisia, antipsychotic induced (alternative agent) (off-label use): Oral (immediate release): Initial: 0.5 to 1 mg twice daily; may increase dose based on response and tolerability up to 6 mg/day, in divided doses (Ref).

Anxiety

Anxiety:

Anxiety and agitation, acute/severe (monotherapy or adjunctive therapy): IM, IV, Oral (immediate release): 0.5 to 2 mg every 4 to 6 hours as needed up to 10 mg/day; adjust dose based on response and tolerability. In severely agitated inpatients, some experts recommend doses up to 4 mg and repeat IM or IV doses as frequently as 10 to 30 minutes; may give alone or in combination with an antipsychotic (Ref).

Anxiety disorders (adjunctive therapy or monotherapy) (alternative agent): Note: Generally used short-term for symptom relief until preferred therapy (eg, serotonin reuptake inhibitor) is effective (eg, 4 to 6 weeks followed by tapering). Long-term low-dose (eg, 0.5 mg/day) therapy may be considered in select patients, when preferred treatments are ineffective or poorly tolerated (Ref). Use with caution in patients with comorbid posttraumatic stress disorder (PTSD); benzodiazepines may cause cognitive changes (Ref).

Immediate release: Oral: Initial: 0.5 to 1 mg 2 to 3 times daily; although most patients will experience relief with this dose, may increase daily dose gradually based on response and tolerability in increments of 1 mg every 2 to 3 days up to 6 mg/day in 2 to 4 divided doses; some patients may require doses up to 10 mg/day for optimal response. Some experts recommend a lower initial dose of 0.5 to 1 mg once to twice daily (Ref).

Extended release 24-hour capsules: Oral: Initial: Determine stable daily dose using IR tablets in 3 divided doses. Maintenance: May convert to extended release at the total daily dose of immediate release; administer once daily in the morning. **Note:** For dosage adjustments, discontinue extended release and switch to immediate release.

Advanced cancer and/or palliative care: IM, IV, Oral (immediate release): 0.5 to 1 mg every 6 to 12 hours as needed (Ref). **Note:** The injectable solution may be administered rectally or subcutaneously, and the tablet and oral solution may be administered sublingually at the same doses when other routes are unavailable (Ref).

Performance- or phobia-related anxiety (monotherapy or adjunctive therapy): **Note:** Provide a test dose, at the same dose to be used for treatment, in advance of the stimulus to ensure tolerability (Ref).

Oral (immediate release): 0.5 to 2 mg once 30 to 60 minutes before the stimulus (Ref).

Procedural anxiety (premedication):

Oral (immediate release), Sublingual (off-label use): 0.5 to 2 mg once 30 to 90 minutes before procedure; if needed due to incomplete response, may repeat the dose (usually at 50% of the initial dose) after 30 to 60 minutes (Ref).

IV: 1 to 4 mg **or** 0.02 to 0.04 mg/kg (maximum single dose: 4 mg) once 5 to 20 minutes before procedure; if needed based on incomplete response and/or duration of procedure, may repeat the dose (usually at 50% of the initial dose) after ≥ 5 minutes (Ref).

Catatonia

Catatonia (off-label use):

Diagnosis: **Note:** Partial, temporary relief of signs following administration is consistent with the diagnosis; a negative response does not rule out catatonia (Ref).

IV (preferred): 1 to 2 mg once; if no response in 5 to 10 minutes, repeat dose once (Ref).

IM: 1 to 2 mg once; may administer up to 2 additional doses at 3-hour intervals if needed (Ref).

Oral (immediate release), Sublingual: 2 mg once; may administer up to 2 additional doses at 3-hour intervals if needed (Ref).

Treatment: **Note:** For patients with malignant catatonia, if rapid resolution is necessary or lorazepam treatment fails, consider electroconvulsive therapy (Ref).

IM, IV, Oral (immediate release): Initial: 1 to 2 mg 3 times daily; IV preferred for initial dosing with switch to oral as patient improves. May increase dose based on response and tolerability in increments of 3 mg every 1 to 2 days to a usual dose of 6 to 24 mg/day (Ref). Doses up to 30 mg/day have been reported (Ref). For patients at risk of cardiorespiratory compromise or oversedation, some experts recommend initiating with 0.5 mg 3 times daily (Ref).

Duration of treatment: Remission is usually achieved in 4 to 10 days; maintenance therapy at the effective dose is usually continued for 3 to 6 months to maintain recovery, although longer courses may be needed (Ref).

Chemotherapy-induced nausea and vomiting, prevention and treatment

Chemotherapy-induced nausea and vomiting, prevention and treatment (adjunctive therapy) (off-label use):

Anticipatory or breakthrough nausea/vomiting, as an adjunct to conventional antiemetics: **Oral (immediate release), IV, Sublingual:** 0.5 to 1 mg every 6 hours as needed; doses up to 2 mg have been described (Ref).

Intoxication

Intoxication: Cocaine, methamphetamine, and other sympathomimetics (off-label use):

Based on limited data: **IV:** 2 to 4 mg every 3 to 10 minutes as needed for agitation, sedation, seizures, hypertension, and tachycardia until desired symptom control achieved. Large cumulative doses may be required for some patients; monitor for respiratory depression and hypotension (Ref). **Note:** Initiating treatment at 1 mg may be adequate in patients who are only mildly or moderately intoxicated, but doses should be repeated or increased as needed. Consider IM administration if IV access is not possible; however, effects will be delayed (Ref).

Mechanically ventilated patients in the ICU, sedation

Mechanically ventilated patients in the ICU, sedation (alternative agent) (off-label use):

Note: Used as part of a multimodal strategy. In general, nonbenzodiazepine sedation is preferred

due to risk of prolonged sedation and delirium with continuous benzodiazepine use. Titrate to light level of sedation (eg, Richmond Agitation-Sedation Scale 0 to –2) or clinical effect (eg, ventilator dyssynchrony). Intermittent as-needed therapy is preferred to avoid drug accumulation and prolonged sedation associated with continuous infusions (Ref). Continuous infusions are not recommended for use in most ICU patients due to propylene glycol (PG) accumulation and subsequent complications (osmol gap metabolic acidosis, kidney failure); monitor PG accumulation with osmol gap; nonbenzodiazepine or midazolam continuous infusions are generally preferred (Ref).

Intermittent (preferred):

Non-weight-based dosing: **IV:** Initial dose: 1 to 4 mg; Maintenance: 1 to 4 mg every 2 to 6 hours as needed (Ref).

Weight-based dosing: **IV:** Initial dose: 0.02 to 0.04 mg/kg (maximum single dose: 4 mg); Maintenance: 0.02 to 0.06 mg/kg every 2 to 6 hours as needed (maximum single dose: 4 mg) (Ref).

Continuous infusion: **IV:** 0.5 to 10 mg/hour **or** 0.01 to 0.1 mg/kg/hour continuous infusion (maximum dose: 10 mg/hour) (Ref).

Neuroleptic malignant syndrome

Neuroleptic malignant syndrome (adjunctive therapy) (off-label use):

Note: Following withdrawal of causative agent while continuing supportive care, may use for muscle rigidity and/or agitation (Ref).

IM, IV: 1 to 2 mg every 4 to 6 hours until symptom resolution (Ref).

Seizures

Seizures: Note: If IV access is not available, IM lorazepam is not recommended due to erratic absorption and a slow time to peak drug levels. May consider alternative routes of other benzodiazepines (eg, IM midazolam) (Ref).

Acute active seizures (non-status epilepticus) (off-label use): IV: 1 to 4 mg given at a maximum rate of 2 mg/minute; may repeat at 3 to 5 minutes if seizures continue (Ref).

Status epilepticus:

Convulsive status epilepticus : **IV, Intraosseous:** 4 mg given at a maximum rate of 2 mg/minute; may repeat at 3 to 5 minutes if seizures continue; a nonbenzodiazepine antiseizure agent should follow to prevent seizure recurrence, even if seizures have ceased (Ref).

Non convulsive status epilepticus: **IV, Intraosseous:** 2 to 4 mg given at a maximum rate of 2 mg/minute; may repeat at 2 to 5 minutes if seizures continue (Ref). Administer in combination with a nonbenzodiazepine antiseizure agent (Ref).

Serotonin syndrome

Serotonin syndrome (serotonin toxicity) (off-label use): IV: 2 to 4 mg IV every 8 to 10 minutes based upon patient response (Ref).

Substance withdrawal

Substance withdrawal:

Alcohol withdrawal syndrome (alternative agent) (off-label use): Note: Withdrawal will progress at different rates in some patients; flexibility in dosing and duration is warranted (Ref). Regimens vary and depend on withdrawal history, degree of current withdrawal symptoms, blood alcohol concentration, and whether the patient is treated inpatient or in the ambulatory setting. Many facilities only treat alcohol withdrawal in the ambulatory setting if Clinical Institute Withdrawal Assessment for Alcohol, revised scale (CIWA-Ar) score is ≤ 15 and there is no history of withdrawal seizures or delirium tremens (Ref). Although longer-acting benzodiazepines are preferred in general over lorazepam due to the lower risk of rebound seizures and withdrawal symptoms, shorter-acting benzodiazepines, including lorazepam, may be preferable in patients with impaired liver function or those who require non-oral treatment (Ref). The following are 2 suggested regimens.

Symptom-triggered regimen:

Note: Some experts recommend symptom-triggered regimens for patients with CIWA-Ar >15 (Ref).

Oral (immediate release), IV: 2 to 4 mg as needed per institution-specific protocol until appropriate sedation achieved; dose and frequency determined by withdrawal symptom severity using a validated severity-assessment scale such as the CIWA-Ar. If frequent treatment is needed, administer IV infusion at a rate of 1 to 4 mg/hour (Ref).

Fixed-dose regimen:

Note: Some experts recommend fixed-dose regimens for patients at risk for mild withdrawal (CIWA-Ar ≤15), particularly in an ambulatory setting where CIWA-Ar scores cannot be reliably administered (Ref). Hold doses for excessive sedation (Ref).

Days 1 through 3: **Oral (immediate release), IV :** 2 mg 3 times daily (Ref).

Day 4: **Oral (immediate release), IV:** 2 mg twice daily, then discontinue lorazepam (Ref).

Breakthrough symptoms: **Oral (immediate release), IV:** May administer additional 1 or 2 mg doses as needed to treat breakthrough symptoms throughout the taper (Ref).

Opioid withdrawal (autonomic instability and agitation) (adjunctive therapy) (alternative agent) (off-label use) : Based on limited data: **IV:** 1 to 2 mg every 10 minutes until hemodynamically stable and adequate sedation (Ref).

Vertigo, acute episodes

Vertigo, acute episodes (alternative agent) (off-label use):

Note: Reserve use for symptomatic relief of episodes lasting several hours to days (maximum duration: 3 days); chronic use may impede adaptation and recovery (Ref).

IM, IV, Oral (immediate release): 0.5 to 2 mg every 4 to 12 hours as needed for up to 48 to 72 hours (Ref).

Discontinuation of therapy: Unless safety concerns require a more rapid withdrawal, gradually taper to detect reemerging symptoms and minimize rebound and withdrawal symptoms in patients receiving therapy ≥4 weeks or as appropriate based on patient-specific factors (Ref).

Low or moderate dose, no concerns for benzodiazepine use disorder: Taper total daily dose by 20% to 25% every week based on response and tolerability (taper increments will be limited by available dosage forms) (Ref).

Extended or high-dose therapy, or suspected benzodiazepine use disorder: For benzodiazepines with half-lives significantly <24 hours, including lorazepam, consider converting to a long-acting benzodiazepine to allow for a more gradual reduction in drug serum concentrations (Ref). Whether converting to a different benzodiazepine or not, assess tolerability of lowering dose by decreasing the total daily dose by 5% to 10% on day 1. If initial decrease is tolerated, begin to taper total daily dose by ~25% every 2 to 4 weeks based on response, tolerability, and individual patient factors (taper increments will be limited by available dosage forms) (Ref). Reduce dose more rapidly in the beginning and slow the dose reduction as the taper progresses because earlier stages of withdrawal are easier to tolerate (Ref). For example, when the dose reaches 25% of the initial dose, some experts suggest tapering more slowly (eg, decreasing the dose every 3 to 6 weeks) (Ref). The optimal duration and taper increment will vary; up to 6 months may be necessary for some patients on higher doses (Ref). In settings where more monitoring is possible (eg, inpatient settings), a faster taper may be appropriate (Ref) .

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A Roberts, PhD, BPharm

(Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: Mild to severe impairment:

Oral: No dosage adjustment necessary (Ref).

Parenteral: No dosage adjustment necessary for acute doses (Ref); use repeated doses with caution; may increase the risk of propylene glycol toxicity. Monitor osmol gap closely as a surrogate marker for propylene glycol accumulation, especially if using for prolonged periods of time or at high doses (Ref).

Hemodialysis, intermittent (thrice weekly): Not significantly dialyzed (Ref): **Oral, Parenteral:** No supplemental dose or dosage adjustment necessary; use repeated parenteral doses with caution; monitor osmol gap closely as a surrogate marker for propylene glycol accumulation, especially if using for prolonged periods of time or at high doses (Ref).

Peritoneal dialysis: Unlikely to be dialyzed (highly protein bound): **Oral, Parenteral:** No dosage adjustment necessary (Ref); use repeated parenteral doses with caution; monitor osmol gap closely as a surrogate marker for propylene glycol accumulation, especially if using for prolonged periods of time or at high doses (Ref).

CRRT: Not significantly dialyzed (Ref): **Oral, Parenteral:** No dosage adjustment necessary (Ref); use repeated intravenous doses with caution; monitor osmol gap closely as a surrogate marker for propylene glycol accumulation, especially if using for prolonged periods of time or at high doses (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Unlikely to be dialyzed (highly protein bound): **Oral, Parenteral:** No dosage adjustment necessary (Ref); use repeated parenteral doses with caution; monitor osmol gap closely as a surrogate marker for propylene glycol accumulation, especially if using for prolonged periods of time or at high doses (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based up on the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Lorazepam metabolism is not altered in patients with cirrhosis (Ref); however, the elimination half-life is prolonged due to increased volume of distribution secondary to decreased protein binding (Ref). First time hepatic encephalopathy events may be precipitated by benzodiazepine use in benzodiazepine-naïve patients with cirrhosis; occurrence was observed primarily between 3 to 10 days of benzodiazepine use in one study, potentially due to drug accumulation (Ref).

Liver impairment prior to treatment initiation:

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Parenteral: Note: Avoid use or use with caution in patients with known hepatic encephalopathy (Ref).

Child-Turcotte-Pugh class A:

Intermittent dosing: No dosage adjustment necessary (Ref).

Continuous infusion: Use should generally be avoided and only if benefits outweigh the risks (Ref).

Child-Turcotte-Pugh class B and C:

Intermittent dosing: No dosage adjustment necessary; however, extending the dosing interval (eg, increasing from every 6 hours to every 12 hours) should be considered when prolonged (>3 days) use is anticipated (Ref).

Continuous infusion: Use should generally be avoided and only if benefits outweigh the risks (Ref).

Oral: Note: ER formulation should not be used in patients with hepatic impairment (Ref).

Child-Turcotte-Pugh class A: No dosage adjustment necessary (Ref).

Child-Turcotte-Pugh class B and C: Note: Use of a shorter acting benzodiazepine (eg, oxazepam) should be considered (Ref).

Initial: Reduce to at least 50% of usual recommended dose and extend dosing interval (eg, if the usual dosing interval is every 6 hours, then extend to every 12 hours) to avoid lorazepam accumulation; gradually titrate based on response and tolerability to the lowest effective dose and frequency. For patients experiencing new onset hepatic encephalopathy, consider dose taper to discontinuation. Avoid use or use with caution in patients with known hepatic encephalopathy (Ref).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, and 3 obesity (BMI ≥ 30 kg/m²):

Intermittent:

IV, IM: Use of non-weight-based dosing is preferred; titrate to clinical effect if needed (Ref). If weight-based doses are utilized, use **adjusted body weight** for initial dose calculations, then titrate to clinical effect if needed (Ref). If rapid sedation or clinical effect is needed, use **actual body weight** for initial weight-based dose calculations, then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, actual body weight to/from adjusted body weight) (Ref). Refer to adult dosing for indication-specific doses.

Oral, Sublingual: Use non-weight-based dosing; titrate to clinical effect if needed (Ref). Refer to adult dosing for indication-specific doses (Ref).

Continuous infusion: Note: Intermittent dosing is preferred for the ICU patient. Use continuous infusion only when ICU patient requiring sedation is refractory to intermittent therapy (Ref). Lorazepam continuous infusions are **not** recommended for use in most ICU patients due to drug accumulation, oversedation, propylene glycol (PG) accumulation, and subsequent complications (osmol gap metabolic acidosis, kidney failure); nonbenzodiazepine or midazolam continuous infusions are generally preferred (Ref). If lorazepam continuous infusion is used, monitor for PG accumulation with osmol gap (Ref).

IV: Use of non-weight-based dosing is preferred; titrate to clinical effect (Ref). If weight-based doses are utilized, use **adjusted body weight** for initial dose calculations, then titrate to clinical effect (Ref). Clinicians should **not** change dosing weight from one weight metric to another during therapy (ie, actual body weight to/from adjusted body weight) (Ref). Refer to adult dosing for indication-specific dosing.

Rationale for recommendations:

There are limited data on the effects of obesity on lorazepam dosing or pharmacokinetics. One lorazepam pharmacokinetic study in subjects with obesity demonstrated both an increased V_d and clearance, but no difference in half-life compared to subjects who were not obese (Ref). Due to the increased V_d , these pharmacokinetic data suggest larger doses may be required. However, larger doses may increase adverse drug effects, particularly in select populations (eg, elderly or nonintubated patients). In general, smaller doses that can be rapidly titrated to effect are preferred (Ref).

Dosing: Older Adult

Note: Avoid use; may be appropriate for seizure disorders, alcohol withdrawal, or severe generalized anxiety disorder (Ref).

Refer to adult dosing. Dose selection should generally be on the low end of the dosage range (initial dose of immediate release not to exceed 2 mg).

Dosing: Pediatric

(For additional information see "Lorazepam: Pediatric drug information")

Agitation, acute; anxiety

Agitation, acute; anxiety:

Note: Due to risk of sedation and respiratory depression, separate parenteral lorazepam and olanzapine doses by at least 1 hour (oral dosing may need longer separation of doses due to delayed onset and peak effect relative to parenteral) (Ref).

Infants and Children <12 years: Limited data available: Oral (immediate release), IV, IM: Usual: 0.05 to 0.1 mg/kg/dose (maximum dose: 2 mg/dose) every 4 to 8 hours (Ref).

Children ≥12 years and Adolescents: Limited data available (weight-based dosing): Oral (immediate release), IV, IM: 0.05 to 0.1 mg/kg/dose 2 or 3 times daily; maximum dose: 2 mg/dose; usual fixed dose range: 0.25 to 2 mg (Ref).

Chemotherapy-induced nausea and vomiting, anticipatory

Chemotherapy-induced nausea and vomiting, anticipatory: Limited data available: Infants, Children, and Adolescents: Immediate release: Oral: 0.04 to 0.08 mg/kg/dose; maximum dose: 2 mg/dose; administer a dose the night before chemotherapy and again the next day prior to chemotherapy administration (Ref).

Sedation, mechanically ventilated patient

Sedation, mechanically ventilated patient: Limited data available:

Infants, Children, and Adolescents: Oral, IV: Initial: 0.05 to 0.1 mg/kg/dose every 4 hours; maximum initial dose: 4 mg/dose; titrate to desired effect (Ref).

Sedation; procedural

Sedation; procedural: Limited data available: Children and Adolescents: Immediate release: Oral: Usual: 0.05 mg/kg; range reported in literature: 0.02 to 0.09 mg/kg (Ref). **Note:** In adults, the maximum dose is 4 mg/dose (Ref).

Status epilepticus

Status epilepticus: Limited data available:

IV: Infants, Children, and Adolescents: IV: 0.1 mg/kg slow IV; may repeat dose once in 5 to 10 minutes; maximum dose: 4 mg/dose (Ref).

Intranasal: **Note:** Lorazepam is not the preferred agent for intranasal administration, guidelines recommend midazolam as the preferred agent (Ref).

Infants, Children, and Adolescents: Intranasal: 0.1 mg/kg/dose; maximum dose: 5 mg/dose (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Oral: Children ≥12 years and Adolescents: There are no dosage adjustments provided in the manufacturer's labeling; however, some clinicians recommend no dosage adjustments are necessary (Ref).

IV: No dosage adjustment necessary for acute doses; use repeated doses with caution; may increase the risk of propylene glycol toxicity. Monitor closely if using for prolonged periods of time or at high doses. In adults, the osmolar gap has been shown to be a surrogate marker for propylene glycol accumulation (Ref).

Dosing: Liver Impairment: Pediatric

Children ≥12 years and Adolescents: No dosage adjustment necessary. For severe hepatic disease, use with caution; benzodiazepines may worsen hepatic encephalopathy.

Adverse Reactions (Significant): Considerations

Anterograde amnesia

Benzodiazepines, including lorazepam, can impair explicit memory and produce short-term **anterograde amnesia** (ie, memory of information or events after drug administration). Retrograde amnesia (ie, events or information prior to drug administration) is unaffected. However, the magnitude of the amnesic effect from benzodiazepines differs among the various agents, depending on their pharmacokinetic/pharmacodynamic properties, route of administration, and dose. Anterograde amnesia is associated with higher doses, IV administration, and benzodiazepines with faster absorption and higher potency. (Ref). Lorazepam has also been reported to produce an atypical effect on implicit memory, disrupt performance on explicit memory tasks, and impair focused attention (Ref).

Mechanism: Dose-related. Benzodiazepines, including lorazepam, bind to the gamma-aminobutyric acid (GABA)-A receptor, subsequently increasing the frequency of chloride channel opening and producing GABA's inhibitory effect throughout the CNS. The GABA-A receptor complex is composed of five subunits, each with multiple isoforms. Each receptor complex contains a benzodiazepine-binding site, which is classified into several types, based on the alpha subunit isoforms and clinical effects related to each type. The BZ1 receptor type, containing the alpha1 subunit isoform, is highly concentrated in the cortex, thalamus, and cerebellum, and is responsible for benzodiazepine-associated anterograde amnesia (Ref).

Onset: Rapid

IV administration: Following a single dose of IV lorazepam, a delayed onset of amnesic action of 15 to 30 minutes has been observed, with a duration between 4 to 8 hours. The onset latency is also dose-related, with shorter onset for higher lorazepam doses (ie, 15 minutes after 4 mg) and a longer onset for lower doses (ie, 30 minutes after 2 mg) (Ref).

Risk factors:

- Specific benzodiazepine: Benzodiazepines, such as lorazepam, with a short half-life, rapid onset, and administered IV versus oral, are associated with a high propensity for amnesic effects. However, midazolam is considered to have superior amnesic properties when compared to lorazepam (Ref).
- Higher doses (Ref)

CNS effects

Lorazepam can cause significant **drowsiness** and **sedated state**; patients also report having "slowed down thinking" (Ref). In addition, lorazepam has been shown to cause psychomotor impairment by increasing reaction time and decreasing reaction accuracy. Benzodiazepines, in general, are associated with an increased risk of motor vehicle accidents and falls (particularly in older adults) (Ref). Tolerance usually develops to the sedative effects (Ref).

Mechanism: Dose-related (Ref); benzodiazepines, including lorazepam, bind to the gamma-aminobutyric acid (GABA)-A receptor, subsequently increasing the frequency of chloride channel opening and producing GABA's inhibitory effect throughout the CNS. Benzodiazepine-associated sedation are mediated by the alpha 1-containing GABA-A receptors (Ref).

Onset: Rapid

- IV administration: Sedative effects: Due to relatively low lipid solubility, onset of sedation is delayed at 15 to 20 minutes, and may require up to 30 minutes (Ref).
- Oral administration: Sedative effects: 20 to 30 minutes (Ref).

Risk factors:

- Older adults (cognitive effects) (Ref)
- Higher doses
- Route of administration (IV administration is associated with a higher risk of sedation versus oral)

Neurodevelopmental effects in children

Animal studies have shown that prolonged or repeated exposure to medications for anesthesia or sedation cause adverse effects on brain maturation resulting in changes in behavior and learning. Some

human studies have also suggested similar effects, including epidemiological studies in humans that have observed various cognitive and behavioral problems, including neurodevelopmental delay (and related diagnoses), learning disabilities, and attention deficit hyperactivity disorder. However, data are limited, inconclusive, and further studies are needed to fully characterize findings. Based on the potential risk, the FDA warned in 2016 that in neonates, children <3 years, or patients in the third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis) undergoing repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures/critical illness, there may be detrimental effects on the child's or fetus' brain development which may lead to various cognitive and behavioral problems. Relatively short exposure (<3 hours) to sedatives or anesthetics during surgery or procedures is unlikely to adversely affect brain development (Ref).

Mechanism: Unknown; in juvenile animal studies, drugs that potentiate gamma-aminobutyric acid activity and/or block N-methyl-D aspartate receptors for >3 hours demonstrated widespread neuronal and oligodendrocyte cell loss along with alteration in synaptic morphology and neurogenesis (Ref).

Risk factors:

- Neonates, children <3 years, or pregnant patients during their third trimester undergoing procedures lasting >3 hours or multiple procedures (Ref)

Paradoxical reactions

Paradoxical reactions, sometimes referred to as disinhibitory reactions or behavioral disinhibition, have been reported in children, adults, and older adults with benzodiazepine use, particularly in those with risk factors. Reactions are relatively uncommon and have been characterized in a number of ways, including increased talkativeness, excitement, restlessness, hyperactivity, sleep disturbances, hostility, rage, agitation, and/or **aggressive behavior** (Ref). In children, paradoxical reactions to benzodiazepines, particularly midazolam, have been most commonly described as agitation, restlessness, inconsolable crying, screaming, disorientation, and/or excitement (Ref).

Mechanism: Dose-related (potentially, although may also be idiosyncratic); exact mechanism is unclear due to limited evidence (Ref). One hypothesis has suggested that increased GABAergic activity in the brain from benzodiazepines causes a decrease in the restraining influence of the frontal cortex, thereby causing excitement, hostility, and rage. An additional proposed mechanism is that benzodiazepines can reduce serotonergic neurotransmission, subsequently causing aggressive behavior. Another mechanism is that in select patients with genetic variations in GABA-A receptors, there is decreased GABA transmission with benzodiazepines which results in neuronal overexcitation manifested clinically as a paradoxical reaction (Ref).

Onset: Rapid; in case reports involving lorazepam, paradoxical reactions occurred within several hours to 24 hours after the first dose; reactions subsided within 24 to 48 hours of discontinuation (Ref)

Risk factors:

Benzodiazepines, in general:

Age related (extremes of age):

- Children (Ref)
- Older adults (Ref)

Disease or condition related:

- Past history of aggressive behavior or violence (Ref)
- Alcoholism or history of alcohol use (Ref)
- Psychiatric or personality disorders, including affective disorder (Ref)
- Dementia (Ref)

Patient related:

- Genetic predisposition (potential risk factor) (ie, variability in the density of the GABA-benzodiazepine receptors throughout the brain, a persistence of a juvenile pattern of benzodiazepine response in adulthood, or multiple allelic forms of the receptors with varying affinities for benzodiazepines have all been suggested as potentially playing a role) (Ref)

Other potential risk factors: **Note:** Not established risk factors since evidence is limited (Ref)

- Higher doses (likely a risk, but not firmly established) (Ref)
- Parenteral benzodiazepine administration (potential risk) (Ref)
- Specific benzodiazepines (potential risk): Benzodiazepines with a short half-life and/or those considered to have higher potency (eg, alprazolam, clonazepam, lorazepam, triazolam) compared to lower potency benzodiazepines are believed to increase the risk (Ref)

Propylene glycol toxicity

Propylene glycol, a solvent used in many pharmaceutical preparations, is found in parenteral lorazepam. High doses, continuous use, and/or prolonged use of IV lorazepam can cause propylene glycol accumulation and subsequent toxicity in adults and children, especially low birth-weight infants and those with hepatic or kidney insufficiency (Ref). Propylene glycol accumulation and toxicity is primarily manifested as hyperosmolar metabolic acidosis (ie, a high anion gap metabolic acidosis with elevated osmol gap) and toxicity is associated with acute kidney injury, intravascular hemolysis, hypotension, cardiac arrhythmias, seizures, mental status changes, and CNS depression (Ref). The osmol gap is correlated with serum propylene glycol concentrations and can be used as a surrogate marker to monitor for propylene glycol accumulation and potential toxicity (Ref). Pediatric patients do accumulate propylene glycol but less often display laboratory abnormalities compared to adults (Ref).

Mechanism: Dose-related; propylene glycol is oxidized by alcohol dehydrogenase to lactaldehyde, which is metabolized by aldehyde dehydrogenase to lactic acid. The accumulation of lactic acid results in metabolic acidosis, but is usually well tolerated. Lactate in this scenario is not generated from any underlying pathology. The lactate is oxidized to pyruvate and then is metabolized by the normal carbohydrate processes.

Onset: Rapid; propylene glycol accumulation, as evidenced by a hyperosmolar anion gap metabolic acidosis, has been observed as early as 48 hours following high-dose continuous infusions (mean lorazepam dose: 8.1 mg/kg; mean infusion rate: 0.16 mg/kg/hour) (Ref).

Risk factors:

- Duration: Continuous infusions >48 hours in adults (Ref)
- Dose:
 - Neonates: Large amounts of propylene glycol orally, IV (eg, >3,000 mg/day), or topically (Ref)
 - Adult: High continuous doses of IV lorazepam (doses exceeding recommended adult dosage range of 0.1 mg/kg/hour in critically ill patients) (Ref)
- Kidney dysfunction (propylene glycol is excreted unchanged in the urine) (Ref)
- Hepatic dysfunction (Ref)
- Concomitant medications also containing propylene glycol (eg, diazepam, esmolol, nitroglycerin, phenytoin, phenobarbital, sulfamethoxazole/trimethoprim) (Ref)
- Impaired alcohol dehydrogenase enzyme system (eg, young children) (Ref)
- Concomitant disulfiram or metronidazole (Ref)
- Alcohol use disorder (Ref)
- Pregnancy (Ref)

- Propylene glycol concentrations exceeding 25 mg/dL. **Note:** Serum propylene glycol concentrations are difficult to obtain in most clinical settings and test turnaround time is slow; osmol gap is recommended as a surrogate marker (Ref).
- In the absence of other osmotic agents (eg, alcohols), an osmol gap of ≥ 10 was predictive of elevated propylene glycol concentrations; values of ≥ 12 suggest propylene glycol toxicity (Ref).
- Critically ill neonates, especially those weighing < 5 kg (Ref)

Withdrawal syndrome

Therapeutic use of benzodiazepines, including lorazepam, is associated with a **withdrawal syndrome** in children and adults, particularly following abrupt or overly rapid discontinuation. Benzodiazepine-associated withdrawal symptoms can include new withdrawal symptoms or rebound symptoms, both of which are typically transient, short-lasting, and reversible. A persistent post-withdrawal disorder, which can be long-lasting, severe, and potentially irreversible, has also been described. In general, withdrawal symptoms are typically mild and characterized as anxiety, panic attacks, restlessness, insomnia and other sleep disturbances, irritability, poor concentration, confusion, nausea/vomiting, weight loss, tremor, diaphoresis, tachycardia, and muscle pain/stiffness. Severe symptoms such as seizure and psychosis may also rarely occur following abrupt discontinuation. Severe withdrawal may be fatal. Data are limited on persistent post-withdrawal disorders, but cognitive impairment, depression, anxiety, sensory disturbances (eg, tinnitus, paresthesia, skin sensations), motor disturbances (eg, muscle pain, weakness, spasms), and GI disturbances have been described (Ref). There are also rare case reports of new-onset mania occurring after abrupt withdrawal of chronic lorazepam (Ref). Severity, onset, and duration of any benzodiazepine withdrawal syndrome varies based on several factors, such as specific benzodiazepine administered (and its half-life), dose, and duration of use. However, in general, new withdrawal symptoms typically resolve within 2 to 4 weeks and rebound symptoms may last 3 weeks, but persistent post-withdrawal symptoms may last > 6 weeks and take 6 to 12 months to completely resolve, and in some cases, persist for years. (Ref).

Mechanism: Withdrawal; exact mechanisms are complex and unclear, but chronic exposure to benzodiazepines alters GABAergic neurotransmission (up/down regulation of gamma-aminobutyric acid [GABA]-A receptor subunits) and rapid or abrupt withdrawal results in underactivity of inhibitory GABA functions subsequently increasing excitatory nervous activity and likely contributing to symptoms associated with withdrawal (Ref). A role of glutamate receptors, including N-methyl-D aspartate (NMDA)-, alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA)-receptors, and metabotropic glutamate (mGlu) receptors, has also been suggested. In mice and rat studies, antagonists for glutamate receptors have shown the ability to potently suppress the withdrawal signs caused by chronic benzodiazepine administration (Ref).

Onset: Varied; onset of symptoms usually occurs within the first day following abrupt withdrawal of lorazepam and any other short- or intermediate-acting benzodiazepines. Long-acting benzodiazepines (eg, diazepam) are associated with an initial onset of withdrawal symptoms 5 days following abrupt discontinuation (Ref). In general, persistent post-withdrawal disorder associated with psychotropic medications (eg, benzodiazepines) have an onset ranging from 24 hours to 6 weeks following a decrease, discontinuation, or switch (Ref).

Risk factors:

Benzodiazepines, in general:

- Abrupt discontinuation (rather than gradual dosage reduction) of a benzodiazepine used long-term (Ref)
- Higher doses (Ref)
- Chronic dosing (ie, intermittent dosing may reduce the risk of withdrawal symptoms) (Ref).
- Long treatment durations (Ref)
- Withdrawal-associated seizure: Predisposed patients (eg, brain damage, alcohol abuse, history of seizure, or those receiving agents that lower the seizure threshold) (Ref)

- Specific benzodiazepine: High-potency benzodiazepines with short and intermediate half-lives, such as lorazepam, are more frequently associated with withdrawal symptoms; benzodiazepines with a short and intermediate half-life have a higher risk for rebound, withdrawal reactions, and dependence compared to long-acting agents (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Local: Pain at injection site (IM: 1% to 17%; IV: 2%)

Nervous system: Drowsiness, sedated state

1% to 10%:

Cardiovascular: Hypotension ($\leq 2\%$)

Local: Erythema at injection site (2%)

Nervous system: Coma ($\leq 1\%$), confusion ($\leq 1\%$), delirium ($\leq 1\%$), depression ($\leq 1\%$), dizziness (7%), excessive crying ($\leq 1\%$), hallucinations (1%), headache ($\leq 1\%$), restlessness ($\leq 1\%$), stupor ($\leq 1\%$), unsteadiness (3%)

Neuromuscular & skeletal: Asthenia ($\leq 4\%$)

Respiratory: Apnea (1%), hypoventilation ($\leq 1\%$), respiratory failure (2%)

<1%:

Cardiovascular: Hypertension

Endocrine & metabolic: Acidosis

Gastrointestinal: Nausea, sialorrhea, vomiting

Genitourinary: Cystitis

Hematologic & oncologic: Thrombocytopenia

Hepatic: Abnormal hepatic function tests, increased serum alkaline phosphatase

Infection: Infection

Local: Injection site reaction

Nervous system: Abnormal gait, abnormality in thinking, agitation, ataxia, brain edema, chills, myoclonus, seizure

Neuromuscular & skeletal: Tremor

Otic: Hearing loss

Respiratory: Hyperventilation

Frequency not defined:

Dermatologic: Alopecia, skin rash

Endocrine & metabolic: Change in libido, hyponatremia, increased lactate dehydrogenase, SIADH

Gastrointestinal: Changes in appetite, constipation

Genitourinary: Impotence, orgasm disturbance

Hematologic & oncologic: Agranulocytosis, leukopenia, pancytopenia

Hepatic: Increased serum bilirubin, increased serum transaminases, jaundice

Hypersensitivity: Anaphylaxis, hypersensitivity reaction

Nervous system: Disinhibition, disorientation, drug dependence, dysarthria, dysautonomia, euphoria, extrapyramidal reaction, fatigue, hypothermia, memory impairment, sleep apnea (exacerbation), slurred speech, suicidal ideation, suicidal tendencies, vertigo, withdrawal syndrome

Ophthalmic: Visual disturbance

Respiratory: Exacerbation of chronic obstructive pulmonary disease, respiratory depression

Miscellaneous: Paradoxical reaction (Ref)

Postmarketing:

Cardiovascular: Bradycardia, cardiac arrhythmia, cardiac failure, heart block, pericardial effusion, prolonged QT interval on ECG (Ref), tachycardia, ventricular arrhythmia

Endocrine & metabolic: Pheochromocytoma (aggravation)

Gastrointestinal: Gastrointestinal hemorrhage

Genitourinary: Urinary incontinence

Hematologic & oncologic: Disorder of hemostatic components of blood, pulmonary hemorrhage

Hepatic: Hepatotoxicity

Hypersensitivity: Fixed drug eruption (Ref)

Nervous system: Aggressive behavior (Ref), anterograde amnesia (Ref), nervousness, neuroleptic malignant syndrome, paralysis

Ophthalmic: Blurred vision, diplopia (Ref)

Respiratory: Pneumothorax, pulmonary edema, pulmonary hypertension

Miscellaneous: Propylene glycol toxicity (IV) (Ref)

Contraindications

Hypersensitivity to lorazepam, any component of the formulation, or other benzodiazepines (cross-sensitivity with other benzodiazepines may exist); acute narrow-angle glaucoma.

Parenteral: Additional contraindications: Hypersensitivity to polyethylene glycol, propylene glycol, or benzyl alcohol; sleep apnea; intra-arterial injection; use in premature infants; severe respiratory insufficiency (except during mechanical ventilation).

Canadian labeling: Additional contraindications (not in the US labeling): Myasthenia gravis.

Warnings/Precautions

Concerns related to adverse effects:

- Sleep-related activities: Hazardous sleep-related activities, such as sleep-driving, cooking and eating food, and making phone calls while asleep, have been noted with benzodiazepines (Dolder 2008).

Disease-related concerns:

- Bariatric surgery: Presurgical assessment of the indication for use, symptoms, and goals of therapy should be documented to enable postsurgical assessment. Monitor for continued efficacy after bariatric surgery and consider switching to an alternate medication if symptoms worsen.

- Depression: Avoid use in patients with depression because of concerns about worsening mood symptoms, particularly if suicidal risk may be present, except for acute or emergency situations (eg, acute agitation, status epilepticus) (Craske 2022).

- Hepatic impairment: Use with caution in patients with hepatic impairment, insufficiency, and/or encephalopathy. Dose adjustment (lower doses) may be needed. May worsen hepatic encephalopathy.

- Renal impairment: Use with caution in patients with renal impairment.

- Respiratory disease: Reduce dose or avoid use in patients with respiratory disease, including chronic obstructive pulmonary disease or sleep apnea. Benzodiazepines may cause significant respiratory depression.

Concurrent drug therapy issues:

- Concomitant use with opioids: In patients already receiving an opioid analgesic, prescribe a lower initial dose of lorazepam than indicated in the absence of an opioid and titrate based on clinical response. If an opioid is initiated in a patient already taking lorazepam, prescribe a lower initial dose of the opioid and titrate based upon clinical response.
- Flumazenil: Flumazenil may cause withdrawal in patients receiving long-term benzodiazepine therapy.

Special populations:

- Debilitated patients: Use with caution in debilitated patients; initial doses should be at the lower end of dosing range.
- Older adult patients: Older adult patients may be at an increased risk of death with use; risk has been found highest within the first 4 months of use in older adult dementia patients (Jennum 2015; Saarelainen 2018).
- Fall risk: Use with extreme caution in patients who are at risk of falls; benzodiazepines have been associated with falls and traumatic injury (Nelson 1999).

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein-binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling.
- Polyethylene glycol: Parenteral formulation may contain polyethylene glycol. May be associated with toxicity in high-dose and/or longer-term therapy.
- Tartrazine: Some formulations may contain tartrazine (FD&C Yellow No. 5), which may cause allergic-type reactions (including bronchial asthma) in susceptible individuals, particularly those who also have aspirin sensitivity.

Other warnings/precautions:

- Abuse, misuse, and substance use disorder: Counsel patients at increased risk on proper use and monitoring for signs and symptoms of abuse, misuse, and substance use disorder. Institute early treatment or refer patients in whom substance use disorder is suspected. Limit dosages and durations to the minimum required.
- Appropriate use: Does not have analgesic, antidepressant, or antipsychotic properties. Status epilepticus should not be treated with injectable benzodiazepines alone; requires close observation and management and possibly ventilatory support. When used as a component of preanesthesia, monitor for heavy sedation and airway obstruction; equipment necessary to maintain airway and ventilatory support should be available.
- Tolerance: Lorazepam is a short half-life benzodiazepine. Duration of action after a single dose is determined by redistribution rather than metabolism. Tolerance develops to the sedative, hypnotic, and antiseizure effects. It does not develop to the anxiolytic effects (Vinkers 2012). Chronic use of this agent may increase the perioperative benzodiazepine dose needed to achieve desired effect.

Warnings: Additional Pediatric Considerations

Use with caution in neonates, especially in preterm infants; several cases of neurotoxicity and myoclonus (rhythmic myoclonic jerking) have been reported.

Some dosage forms may contain propylene glycol; in neonates, large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Capsule ER 24 Hour Sprinkle, Oral:

Loreev XR: 1 mg [contains corn starch, fd&c yellow #5 (tartrazine)]

Loreev XR: 1.5 mg [contains corn starch, fd&c blue #1 (brilliant blue)]

Loreev XR: 2 mg [contains corn starch, quinoline yellow (d&c yellow #10)]

Loreev XR: 3 mg [contains corn starch, fd&c blue #1 (brilliant blue), fd&c yellow #6 (sunset yellow), quinoline yellow (d&c yellow #10)]

Concentrate, Oral:

LORazepam Intensol: 2 mg/mL (30 mL) [alcohol free, dye free, sugar free; contains polyethylene glycol (macrogol), propylene glycol; unflavored flavor]

Generic: 2 mg/mL (30 mL)

Solution, Injection:

Ativan: 2 mg/mL (1 mL, 10 mL); 4 mg/mL (1 mL, 10 mL) [contains benzyl alcohol, polyethylene glycol (macrogol), propylene glycol]

Generic: 2 mg/mL (1 mL, 10 mL); 4 mg/mL (1 mL, 10 mL)

Tablet, Oral:

Ativan: 0.5 mg

Ativan: 1 mg, 2 mg [scored]

Generic: 0.5 mg, 1 mg, 2 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Capsule ER 24 Hour Sprinkle (Loreev XR Oral)

1 mg (per each): \$17.85

1.5 mg (per each): \$17.85

2 mg (per each): \$17.85

3 mg (per each): \$17.85

Concentrate (LORazepam Intensol Oral)

2 mg/mL (per mL): \$1.60

Concentrate (LORazepam Oral)

2 mg/mL (per mL): \$1.60 - \$10.00

Solution (Ativan Injection)

2 mg/mL (per mL): \$2.88

4 mg/mL (per mL): \$3.17

Solution (LORazepam Injection)

2 mg/mL (per mL): \$1.70 - \$19.20

4 mg/mL (per mL): \$2.16 - \$4.10

Tablets (Ativan Oral)

0.5 mg (per each): \$53.61

1 mg (per each): \$71.62

2 mg (per each): \$114.14

Tablets (LORazepam Oral)

0.5 mg (per each): \$0.09 - \$0.66

1 mg (per each): \$0.10 - \$7.20

2 mg (per each): \$0.14 - \$9.60

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Generic: 2 mg/mL ([DSC]); 4 mg/mL (1 mL)

Tablet, Oral:

Ativan: 0.5 mg [DSC], 1 mg [DSC], 2 mg [DSC]

Generic: 0.5 mg, 1 mg, 2 mg

Tablet Sublingual, Sublingual:

Ativan: 0.5 mg [contains corn starch, fd&c blue #1 (brill blue) aluminum lake, fd&c yellow #6(sunset yellow)alumin lake, quinoline (d&c yellow #10) aluminum lake]

Ativan: 1 mg [contains corn starch]

Ativan: 2 mg [DSC] [contains fd&c blue #2 (indigo carm) aluminum lake]

Generic: 0.5 mg, 1 mg, 2 mg

Extemporaneous Preparations

Note: Commercial oral solution is available (2 mg/mL)

Two different 1 mg/mL oral suspensions may be made from different generic lorazepam tablets (Mylan Pharmaceuticals or Watson Laboratories), sterile water, Ora-Sweet, and Ora-Plus.

Mylan tablets: Place one-hundred-eighty 2 mg tablets in a 12-ounce amber glass bottle; add 144 mL of sterile water to disperse the tablets; shake until slurry is formed. Add 108 mL Ora-Plus in

incremental proportions; then add a quantity of Ora-Sweet sufficient to make 360 mL. Label "shake well" and "refrigerate". Stable for 91 days when stored in amber glass prescription bottles at room temperature or refrigerated (preferred).

Watson tablets: Place one-hundred-eighty 2 mg tablets in a 12-ounce amber glass bottle; add 48 mL sterile water to disperse the tablets; shake until slurry is formed. Add 156 mL of Ora-Plus in incremental proportions; then add a quantity of Ora-Sweet sufficient to make 360 mL. Label "shake well" and "refrigerate". Store in amber glass prescription bottles. Stable for 63 days at room temperature or 91 days refrigerated (Ref).

Controlled Substance

C-IV

Administration: Adult

IM: Should be administered (undiluted) deep into the muscle mass.

Intraosseous: Status epilepticus: Intraosseous administration is a reasonable alternative when quick IV access is not feasible (Ref).

IV injection: Dilute prior to use (according to the manufacturer); continuous infusion may also be administered undiluted (up to 4 mg/mL) into a central vein or into a peripheral vein with a compatible IV solution infusing concurrently at a high flow rate (Ref). Do not exceed 2 mg/minute or 0.05 mg/kg over 2 to 5 minutes. Monitor IV site during administration. Avoid intra-arterial administration. Avoid extravasation.

Oral:

ER capsule: Administer with or without food. Do not crush or chew. Swallow whole or open and sprinkle the entire contents of capsule over a tablespoon of applesauce, then drink water after consuming the applesauce (without chewing). Consume entire contents of capsule within 2 hours of opening capsule.

Bariatric surgery: Lorazepam is available as an ER formulation and the release characteristics may be significantly altered in an unknown manner in patients who have undergone bariatric surgery, including when the sprinkle capsules are opened for administration; providers should determine if the condition being treated can be safely monitored or if a switch to an alternative formulation is necessary (Ref). Lorazepam is also available in an IR formulation.

Oral concentrate: Use only the provided calibrated dropper to withdraw the prescribed dose. Mix the dose with liquid (eg, water, juice, soda, soda-like beverage) or semisolid food (eg, applesauce, pudding), and stir for a few seconds to blend completely. The prepared mixture should be administered immediately.

SubQ (off-label route): Some experts administer the injectable solution subcutaneously when other routes are unavailable (eg, comfort care settings) (Ref).

Rectal (off-label route): Injectable solution may be administered rectally when other routes are unavailable (eg, comfort care settings) (Ref).

Sublingual:

Oral concentrate (off-label route): May be administered sublingually when other routes are unavailable (eg, comfort care settings) (Ref).

Oral tablet (off-label route): May be administered sublingually when other routes are unavailable (eg, comfort care settings) (Ref).

Sublingual tablet [Canadian product]: Place under tongue; patient should not swallow for at least 2 minutes.

Preparation for Administration: Adult

IV injection: According to the manufacturer, dilute IV dose prior to use with an equal volume of compatible diluent (D5W, NS, SWFI).

Infusion: Precipitation may occur upon dilution when preparing an infusion. Use 2 mg/mL injectable vial to prepare; there may be decreased stability when using 4 mg/mL vial. Dilute to ≤ 1 mg/mL with a compatible diluent in a non-PVC (eg, polyolefin, glass) container (consult parenteral admixture resource for additional detailed recommendations). Can also be administered undiluted (up to 4 mg/mL) via infusion into a central vein or into a peripheral vein with a running compatible maintenance IV solution (Ref).

IM: Administer undiluted.

Administration: Pediatric

Oral:

Tablet: May administer with food to decrease GI distress.

Oral concentrate solution: Measure dose using a calibrated oral dosing syringe (or other accurate dose-measuring device). Dilute oral solution in water, juice, or soda, or add to semisolid food (eg, applesauce, pudding); stir gently for a few seconds. Administer dose immediately; do not store for future use.

Intranasal: Administer undiluted (injectable formulation) into one nostril using a needleless syringe or nasal atomizer (Ref).

Parenteral:

IV: Dilute prior to administration. Do not exceed 2 mg/minute or 0.05 mg/kg over 2 to 5 minutes; administer IV using repeated aspiration with slow IV injection, to make sure the injection is not intra-arterial and that perivascular extravasation has not occurred.

IM: Administer undiluted by deep injection into muscle mass

Preparation for Administration: Pediatric

Parenteral:

IV: Dilute dose prior to use with an equal volume of compatible diluent (D5W, NS, SWFI).

Neonates: Since both concentrations of the injection contain 2% benzyl alcohol, some experts recommend using the 4 mg/mL product for dilution with preservative free SWFI to make a 0.4 mg/mL dilution (in order to decrease the amount of benzyl alcohol delivered to the neonate); however, the stability of this dilution has not been studied.

Storage/Stability

Parenteral: Intact vials should be refrigerated (room temperature storage information may be available; contact product manufacturer to obtain current recommendations). Protect from light. Storage of solution diluted for infusion information is not provided in the manufacturer's labeling; refer to individual institutions policies and procedures.

Oral concentrate: Store at 2°C to 8°C (36°F to 46°F); protect from light. Dispense only in original bottle and only with manufacturer-supplied calibrated dropper; discard open bottle after 90 days.

Oral ER capsule: Store at 20°C to 25°C (68°F to 77°F); excursions are permitted between 15°C and 30°C (59°F and 86°F).

Oral IR tablet: Store at 25°C (77°F); excursions are permitted between 15°C and 30°C (59°F and 86°F).

Sublingual tablet [Canadian product]: Store at 15°C to 25°C (59°F to 77°F). Protect from light.

Use: Labeled Indications

Anxiety (oral):

Extended release: Treatment of anxiety disorders in adults who are receiving stable, evenly divided, 3 times daily dosing with lorazepam immediate release.

Immediate release: Management of anxiety disorders or short-term (≤ 4 months) relief of anxiety.

Procedural anxiety, premedication (injection): Anesthesia premedication in adults to relieve anxiety or to produce amnesia (diminish recall) or sedation.

Status epilepticus (injection): Treatment of status epilepticus. May be used off label for acute seizures that have not yet progressed to status epilepticus.

Use: Off-Label: Adult

Akathisia, antipsychotic-induced; Alcohol withdrawal syndrome; Catatonia; Chemotherapy-induced nausea and vomiting; Intoxication: Cocaine, methamphetamine, and other sympathomimetics; Mechanically ventilated patients in the ICU, sedation; Neuroleptic malignant syndrome; Opioid withdrawal; Serotonin syndrome (serotonin toxicity); Vertigo, acute episodes

Medication Safety Issues

Sound-alike/look-alike issues:

LORazepam may be confused with ALPRAZolam, clonazepam, diazepam, Klonopin, Lovaza, temazepam, zolpidem.

Ativan may be confused with Ambien, Atarax, Atgam, Avitene.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (moderate sedation agent, IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Older Adult: High-Risk Medication:

Beers Criteria: Lorazepam is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older because of risk of abuse, misuse, physical dependence, and addiction. In addition, older adults have increased risk of impaired cognition, delirium, falls, fractures, and motor vehicle accidents with benzodiazepine use. However, benzodiazepines may be appropriate in the elderly when used for seizure disorders, rapid eye movement sleep behavior disorder, benzodiazepine or ethanol withdrawal, severe generalized anxiety disorder, or periprocedural anesthesia (Beers Criteria [AGS 2023]).

Lorazepam is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥ 65 years of age) with a history of recurrent falls due to an increased risk of falls. Use of benzodiazepines is not recommended for ≥ 4 weeks; limit insomnia treatment to < 2 weeks. In addition, some disease states of concern include dementia and respiratory failure (acute or chronic) (O'Mahony 2023).

Administration issues:

Injection dosage form contains propylene glycol. Monitor for toxicity when administering continuous or high-dose lorazepam infusions.

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Alcohol (Ethyl): May increase CNS depressant effects of LORazepam. Alcohol (Ethyl) may increase serum concentrations of LORazepam. Specifically, this increase in concentration would only occur with use of lorazepam extended release capsules and alcohol. *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylinoxide: CNS Depressants may increase CNS depressant effects of Amitriptylinoxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ARIPrazole Lauroxil: May increase CNS depressant effects of Benzodiazepines. ARIPrazole Lauroxil may increase hypotensive effects of Benzodiazepines. Specifically, the risk of orthostatic hypotension may be increased. *Risk C: Monitor*

ARIPrazole: May increase CNS depressant effects of Benzodiazepines. ARIPrazole may increase hypotensive effects of Benzodiazepines. Specifically, orthostatic hypotension may be increased. *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benzodiazepines: CNS Depressants may increase CNS depressant effects of Benzodiazepines. *Risk C: Monitor*

Beta-Acetyldigoxin: Benzodiazepines may increase serum concentrations of Beta-Acetyldigoxin. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buclicline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cannabinoid-Containing Products: CNS Depressants may increase CNS depressant effects of Cannabinoid-Containing Products. *Risk C: Monitor*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Certoparin: May increase serum concentrations of Benzodiazepines. *Risk C: Monitor*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Chlorpheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: Benzodiazepines may increase adverse/toxic effects of CloZAPine. *Risk C: Monitor*

CloZAPine: CNS Depressants may increase CNS depressant effects of CloZAPine. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Benzodiazepines. *Risk C: Monitor*

Corticosteroids (Orally Inhaled): Benzodiazepines may increase adverse/toxic effects of Corticosteroids (Orally Inhaled). Specifically, the risk of pneumonia may be increased. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Dantrolene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

Dexmedetomidine: CNS Depressants may increase CNS depressant effects of Dexmedetomidine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimenhydrinate: CNS depressant effects of CNS Depressants and Dimenhydrinate may increase with coadministration. *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Diphenhydramine (Systemic): CNS depressant effects of CNS Depressants and Diphenhydramine (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Disulfiram: May increase adverse/toxic effects of Products Containing Propylene Glycol. A disulfiram reaction may occur. Management: Consider avoiding concomitant use of disulfiram and products that contain propylene glycol. Note that not all preparations or dosage forms of a specific drug will contain propylene glycol. *Risk D: Consider Therapy Modification*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

FluPHENAZine: CNS Depressants may increase CNS depressant effects of FluPHENAZine. *Risk C: Monitor*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

HydrOXYzine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Ilaprazole: May increase serum concentrations of Benzodiazepines. *Risk C: Monitor*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketamine: CNS Depressants may increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRAcetam: CNS Depressants may increase CNS depressant effects of LevETIRAcetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Loxapine: May increase adverse/toxic effects of LORazepam. Specifically, prolonged stupor, respiratory depression, and/or hypotension. *Risk C: Monitor*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Melatonin: May increase sedative effects of Benzodiazepines. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methadone: Benzodiazepines may increase CNS depressant effects of Methadone. Management: Clinicians should generally avoid concurrent use of methadone and benzodiazepines when possible; any combined use should be undertaken with extra caution. *Risk D: Consider Therapy Modification*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methoxyflurane: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: Benzodiazepines may increase CNS depressant effects of Mirtazapine. *Risk X: Avoid*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

OLANZapine: Benzodiazepines may increase adverse/toxic effects of OLANZapine. Management: Monitor closely for hypotension, respiratory or central nervous system depression, and bradycardia if olanzapine is combined with benzodiazepines. Use of parenteral benzodiazepines with IM olanzapine is not recommended. *Risk C: Monitor*

OLANZapine: CNS depressant effects of OLANZapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): Benzodiazepines may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). *Risk X: Avoid*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Pimozide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Probenecid: May increase serum concentrations of LORazepam. Management: Reduce lorazepam dose 50% during coadministration with probenecid. Monitor for increased and prolonged lorazepam effects, particularly CNS depressant effects. Patients using lorazepam ER capsules should be switched to lorazepam tablets. *Risk D: Consider Therapy Modification*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propofol: CNS Depressants may increase CNS depressant effects of Propofol. *Risk C: Monitor*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pyrimethamine: LORazepam may increase hepatotoxic effects of Pyrimethamine. *Risk C: Monitor*

QUetiapine: CNS Depressants may increase CNS depressant effects of QUetiapine. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

RisperiDONE: CNS depressant effects of CNS Depressants and RisperiDONE may increase with coadministration. *Risk C: Monitor*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIrole: CNS Depressants may increase sedative effects of ROPINIrole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Secnidazole: Products Containing Propylene Glycol may increase adverse/toxic effects of Secnidazole. *Risk X: Avoid*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Teduglutide: May increase serum concentrations of Benzodiazepines. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Theophylline Derivatives: May decrease therapeutic effects of Benzodiazepines. *Risk C: Monitor*

Thioridazine: CNS Depressants may increase CNS depressant effects of Thioridazine. *Risk C: Monitor*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

TiaGABine: CNS Depressants may increase CNS depressant effects of TiaGABine. *Risk C: Monitor*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Tripolidine: CNS Depressants may increase CNS depressant effects of Tripolidine. *Risk C: Monitor*

UGT2B15 Inhibitors: May increase serum concentrations of LORazepam. Management: Avoid coadministration of UGT2B15 inhibitors and extended release lorazepam capsules. If coadministration is required, discontinue lorazepam extended release capsules and use lorazepam tablets instead. Monitor for increased lorazepam toxicities. *Risk D: Consider Therapy Modification*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: May increase serum concentrations of LORazepam. CNS depressant effects of LORazepam and Valproic Acid and Derivatives may increase with coadministration. Management: Reduce lorazepam dose 50% during coadministration with valproate products. Monitor for increased lorazepam effects, particularly CNS depression. Patients taking lorazepam extended-release capsules should be switched to lorazepam tablets. *Risk D: Consider Therapy Modification*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Yohimbine: May decrease therapeutic effects of Antianxiety Agents. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Ziprasidone: CNS Depressants may increase CNS depressant effects of Ziprasidone. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Reproductive Considerations

Evaluate pregnancy status prior to use. Pregnancy testing is recommended before treating acute alcohol withdrawal symptoms; however, treatment should not be delayed in symptomatic patients if pregnancy status is unknown (ASAM 2020).

Lorazepam is not a first-line medication to treat anxiety disorders prior to conception in patients who are treatment naive or who do not have a history of effective treatment with a specific medication. Patients effectively treated may continue their current medication when planning a pregnancy unless contraindications exist. Management of mental health conditions in patients who could become pregnant should be based on a shared decision-making process that considers the possibility of pregnancy during treatment (ACOG 2023; BAP [McAllister-Williams 2017]).

Pregnancy Considerations

Lorazepam and its metabolite cross the human placenta (Kanto 1982; Papini 2006).

In utero exposure to benzodiazepines has the potential to cause harm to the fetus. Teratogenic effects have been observed in some studies; however, a clear association has not been reported and additional data are needed (Bellantuono 2013; Chuang 2024; Freeman 2018; Grigoriadis 2019; Tinker 2019; Wu 2024). Exposure to a benzodiazepine late in pregnancy may cause neonatal sedation (hypotonia, lethargy, respiratory depression) and/or symptoms of neonatal withdrawal (feeding difficulties, hyperreflexia, inconsolable crying, irritability, restlessness, tremors). Monitor newborns exposed to lorazepam in utero for adverse events. Data related to long-term effects on neurodevelopment following maternal use of benzodiazepines are inconclusive (Andrade 2024; Radojčić 2017; Sundbakk 2025; Wang 2022).

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of lorazepam may be altered, resulting in an increased apparent V_d , increased apparent oral clearance, and decreased elimination half-life (Papini 2006). Elimination of lorazepam in the newborn infant is slow; following in utero exposure, term infants may excrete lorazepam for at least 8 days (Whitelaw 1981).

Benzodiazepines are one of the preferred options for treating pregnant patients with alcohol withdrawal. Inpatient treatment is recommended during pregnancy. Agents with a shorter onset and

duration of action should be used if near delivery. Monitor newborns for fetal alcohol spectrum disorder, sedative withdrawal, and intoxication (ASAM 2020).

Lorazepam is recommended for the treatment of postpartum psychosis (ACOG 2023).

Untreated and undertreated mental health conditions are associated with adverse pregnancy outcomes. Anxiety disorders during pregnancy are associated with low birth weight, preterm birth, and adverse behavioral outcomes in the offspring. Discontinuing effective medications during pregnancy increases the risk of relapse. Management should be made as part of a shared decision-making process. Agents other than lorazepam may be preferred when treatment for anxiety is initiated for the first time during pregnancy. The use of benzodiazepines for the treatment of perinatal anxiety should be avoided or used sparingly until preferred therapy is initiated and effective, or when preferred therapy is not effective; lorazepam may be a preferred benzodiazepine in patients with preeclampsia. When medications are used, the lowest effective dose of a single agent is recommended. Optimize dosing prior to changing a medication or adding additional agents whenever possible. Monthly monitoring for symptom improvement with a validated screening tool during pregnancy is recommended. Manage side effects as needed (ACOG 2023).

Treatment of status epilepticus during pregnancy should consider GA and etiology of seizures (eg, eclampsia versus other causes) (NCS [Brophy 2012]; Rajiv 2019)

Data collection to monitor pregnancy and infant outcomes is ongoing. Encourage pregnant patients to enroll in the National Pregnancy Registry for Psychiatric Medications (1-866-961-2388 or <https://womensmentalhealth.org/research/pregnancyregistry/>).

Breastfeeding Considerations

Lorazepam is present in breast milk.

Data related to the presence of lorazepam in breast milk are available from multiple sources:

- Following maternal administration of oral lorazepam 2.5 mg twice daily for the first 5 days postpartum, breast milk concentrations of lorazepam and lorazepam glucuronide were 12 ng/mL and 35 ng/mL, respectively. Authors of the study calculated the estimated exposure of lorazepam to this breastfeeding infant to be 2.6 mcg/kg/day (based on a milk intake of 200 mL/kg/day). The authors of the study also calculated the estimated exposure of lorazepam + lorazepam glucuronide via breast milk to be 9.6 mcg/kg/day. Adverse events were not observed in the neonate (Whitelaw 1981).
- Breast milk from 4 lactating patients was collected following a single dose of lorazepam 3.5 mg given orally prior to postpartum sterilization. Four hours after the dose, concentrations of lorazepam in breast milk were 8 to 9 ng/mL (Summerfield 1985).
- Breast milk was collected from 3 patients, 3 to 6 days postpartum, who had been taking lorazepam 0.5 mg/day during pregnancy and after delivery. Breast milk was sampled 2 hours after the maternal dose, approximately when the peak maternal plasma concentration occurs. Additional breast milk samples were collected at random times during the 1-month check-up. Peak lorazepam concentrations in breast milk ranged from 1.64 to 1.98 ng/mL when sampled postpartum. The authors of the study calculated the relative infant dose (RID) of lorazepam via breast milk, compared to the weight-adjusted maternal dose, 2 different ways - first by using the actual amount of breast milk ingested, then by using a theoretical milk intake of 150 mL/kg/day (the average amount ingested by an exclusively breastfed infant). The RID of lorazepam for these patients was calculated to be 1% to 1.5% (based on the actual amount of breast milk ingested) and 2.1% to 4.4% (based on theoretical breast milk intake). When breast milk was sampled at the 1-month visit, the RID for lorazepam ranged from 0.7% to 3.1% (based on actual intake) and 1.7% to 2.7% (based on theoretical value) (Nishimura 2021).
- Higher breast milk concentrations were observed in 1 mother who received both oral lorazepam and lormetazepam, which is partially metabolized to lorazepam (Lemmer 2007).
- In general, breastfeeding is considered acceptable when the RID is <10% (Anderson 2016; Ito 2000); however, some sources note breastfeeding should only be considered if the RID is <5%

for psychotropic agents (Eleftheriou 2024).

Drowsiness, lethargy, or weight loss in breastfeeding infants have been observed in case reports following maternal use of some benzodiazepines (Iqbal 2002). Adverse events following benzodiazepine exposure via breast milk were evaluated in a study of 124 infants (2 to 24 months of age). Lorazepam use was reported by 54% of their mothers. CNS depression was observed in 2 breastfed infants, neither exposed to lorazepam (Kelly 2012).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. A benzodiazepine with a short half-life (such as lorazepam) can be considered for the treatment of anxiety in patients who are breastfeeding (Eleftheriou 2024).

Monitoring Parameters

Respiratory and cardiovascular status, BP, heart rate, symptoms of anxiety, mental alertness.

Long-term therapy: CBC, liver function tests, LDH.

High-dose or continuous IV use or IV use in patients with renal impairment: Clinical signs of propylene glycol toxicity, serum creatinine, BUN, serum lactate, osmol gap; **Note:** An osmol gap of ≥ 10 was predictive of elevated propylene glycol concentrations; values of ≥ 12 suggest propylene glycol toxicity (Arroliga 2004; Barnes 2006; Yahwak 2008).

Critically ill patients: Assess and adjust sedation according to scoring system (Richmond Agitation-Sedation Scale [RASS] or Sedation-Agitation Scale [SAS]) (SCCM [Devlin 2018]).

Mechanism of Action

Short-to-intermediate-acting benzodiazepine (based on half-life) (Griffin 2013). Binds to stereospecific benzodiazepine receptors on the postsynaptic GABA neuron at several sites within the central nervous system, including the limbic system, reticular formation. Enhancement of the inhibitory effect of GABA on neuronal excitability results by increased neuronal membrane permeability to chloride ions. This shift in chloride ions results in hyperpolarization (a less excitable state) and stabilization. Benzodiazepine receptors and effects appear to be linked to the GABA-A receptors. Benzodiazepines do not bind to GABA-B receptors.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Antiseizure medication: IV: Within 10 minutes.

Hypnosis: IM: 20 to 30 minutes.

Sedation:

IV: 15 to 20 minutes (dose dependent) (Barr 2013; Horn 2004).

Oral: 20 to 30 minutes (Horn 2004).

Duration:

Adults: Classified as a short-to-intermediate-acting benzodiazepine; classification based on benzodiazepines with half-life of < 40 hours (Griffin 2013).

Indication specific duration:

Anesthesia premedication: Adults: IM, IV: ~ 6 to 8 hours.

Absorption: IM: Delayed relative to IV (Leppik 2015; Wermeling 2001); Oral: Readily absorbed.

Distribution:

Neonates (GA ≥ 37 weeks): IV: $V_d: 0.76 \pm 0.37$ L/kg (range: 0.14 to 1.3 L/kg) (McDermott 1992).

Pediatric patients (Chamberlain 2012): IV: $V_d:$

5 months to < 3 years: 1.62 L/kg (range: 0.67 to 3.4 L/kg).

3 to <13 years: 1.5 L/kg (range: 0.49 to 3 L/kg).

13 to <18 years: 1.27 L/kg (range: 1 to 1.54 L/kg).

Adults:

IV: V_d : 1.3 L/kg.

Oral: Extended release: V_d : 117 L.

Protein binding: 85% to 91%.

Metabolism: Hepatic; rapidly conjugated to lorazepam glucuronide (inactive).

Bioavailability: Oral: 90%.

Half-life elimination:

Full-term neonates: IV: 40.2 ± 16.5 hours; range: 18 to 73 hours (McDermott 1992).

Pediatric patients (Chamberlain 2012): IV:

5 months to <3 years: 15.8 hours (range: 5.9 to 28.4 hours).

3 to <13 years: 16.9 hours (range: 7.5 to 40.6 hours).

13 to <18 years: 17.8 hours (range: 8.2 to 42 hours).

Adults: Oral: Extended release: $\sim 20.2 \pm 7.2$ hours; Immediate release: ~ 12 hours; IV: ~ 14 hours; IM: ~ 13 to 18 hours (Greenblatt 1983); End-stage renal disease (ESRD): ~ 18 hours.

Time to peak:

IM: ≤ 3 hours.

Oral:

Extended release: 14 hours (range: 7 to 24 hours).

Immediate release: ~ 2 hours.

Sublingual tablet [Canadian product]: 1 hour.

Excretion: Urine ($\sim 88\%$; predominantly as inactive metabolites); feces ($\sim 7\%$).

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and as follows, must be dispensed with this medication:

Ativan: https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/017794s049lbl.pdf#page=14

Loreev XR: https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/214826s003lbl.pdf#page=16

Patient Counseling Points

(For additional information see "Lorazepam: Patient drug information")

What is this drug used for?

- It is used to treat anxiety.
- It is used to treat seizures.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

- Feeling dizzy, sleepy, tired, or weak

- Headache

Injection:

- Irritation where the shot is given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- New or worse behavior or mood changes like depression or thoughts of suicide
- Hallucinations (seeing or hearing things that are not there)
- Change in balance
- Feeling confused
- Feeling nervous and excitable
- Restlessness
- Trouble sleeping
- Sex problems
- Memory problems or loss
- Severe dizziness or passing out
- Change in eyesight
- Muscle weakness
- Dark urine or yellow skin or eyes
- Trouble breathing, slow breathing, or shallow breathing
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Apo lorazepam | Ativan;

- (AR) Argentina: Aplacasse | Calmatron | Emotival | Kalmalin | Lorazepam fabra | Lorazepam klonal | Lorazepam northia | Lorazepam richet | Lorazepam vannier | Lorazepam | Lorazepam Chobet | Lorazepam lacefa | Lorezan | Microzepam | Nervistop | Nervistop | lacefa | Plardox | Secodal | Sedatival | Sidenar | Trapax | Trapecis | Tratenamin;
- (AT) Austria: Merlit | Temesta;
- (AU) Australia: Apo lorazepam | Ativan | Bioplus sedacalm | Lorazepam lupin | Lorazepam mylan;
- (BD) Bangladesh: Lozicum | Trapex;
- (BE) Belgium: Calmamed | Docloraze | Lauracalm | Lorazemed | Lorazepam efeka | Lorazepam Eurogenerics | Lorazepam mylan | Lorazepam ratiopharm | Lorazepam teva generics belgium | Lorazetop | Loridem | Serenase | Temesta | Vigiten;
- (BF) Burkina Faso: Temesta;
- (BG) Bulgaria: Lorapam;
- (BR) Brazil: Lorapan | Lorax | Lorazefast | Max-pax | Mesmerin;
- (CH) Switzerland: Ativan | Lorasifar | Sedazin | Temesta;
- (CL) Chile: Abinol | Amparax;
- (CN) China: Jia pu le | Le la an | Lora;
- (CO) Colombia: Ativan | Lorazepam MK | Lorazepam;
- (CR) Costa Rica: Lorazepam MK;
- (CZ) Czech Republic: Apo lorazepam | Tavor | Temesta;
- (DE) Germany: Duralozam | Laubeel | Lorazepam aristo | Lorazepam CT | Lorazepam Dura | Punktyl | Somagerol | Tavor | Tavor expidet | Tolid;
- (DO) Dominican Republic: Ativan | Lorep | Placinoral | Sedarkey | Sonamin | Tensopan;
- (EC) Ecuador: Ativan;
- (EE) Estonia: Laubeel | Lorans | Temesta;
- (EG) Egypt: Apo lorazepam | Ativan;
- (ES) Spain: Donix | Idalprem | Lorazepam aristo | Lorazepam Cinfa | Lorazepam desgen | Lorazepam Kern pharma | Lorazepam normon | Lorazepam pensa | Lorazepam vir | Orfidal | Sedicepan;
- (FI) Finland: Lorazepam orifarm | Lorazepam orion | Temesta | Vanapam;
- (FR) France: Equitam | Lorazepam biogaran | Lorazepam merck | Temesta;
- (GB) United Kingdom: Abbloraz | Almazine | Ativan | Lorazepam aps | Lorazepam arrow | Lorazepam astec | Lorazepam berk | Lorazepam cox | Lorazepam dc | Lorazepam kent | Lorazepam sandoz | Lorazepam steinhard;
- (GR) Greece: Aripax | Dorm | Modium | Nifalin | Novhepar | Proneurit | Tavor | Titus | Trankilium;
- (HK) Hong Kong: Ativan | Lora | Lorans | Lorivan | Lotevem | Lozicum | Pms lorazepam | Silence;
- (HR) Croatia: Lorsilan;
- (HU) Hungary: Loranxil | Lorazepam Aramis | Rilex;
- (ID) Indonesia: Ativan | Merlopam | Renaquil;
- (IE) Ireland: Ativan | Azepam;
- (IL) Israel: Lorivan;
- (IN) India: Almazine | Anxilor | Anxipose | Aticool | Atipam | Ativan | Axitan | Benj | Calmese | Emelor | Jerocen | L pam | L zepam | Larox | Larpose | Lopez | Loracalm | Loranix | Loranza | Lorat | Lorawel | Lorax | Loraz | Lorazen | Lorazin | Lorcin | Lored | Lorel | Lorepa | Loricon | Loripam | Lorivan | Lormac | Lorvan | Loxum | Lozep 2 | Meripose | Neulora | Norpose | Orazep | Pezlor | Proline | Psypose | Trapex | Zepnap;
- (IT) Italy: Control | Lorans | Lorazepam abc | Lorazepam actavis | Lorazepam almus | Lorazepam alter | Lorazepam aristo | Lorazepam doc generici | Lorazepam eg | Lorazepam mebel | Lorazepam ranbaxy | Lorazepam ratiopharm | Lorazepam reddy | Lorxagen | Tavor | Zeloram;
- (JO) Jordan: Ativan | Lorans;
- (JP) Japan: Azrogen | Lora pita | Lorazepam sawai | Lorazepam towa | Rocosgen | Symplex | U pan | Wypax | Wypax yamanouchi;
- (KR) Korea, Republic of: Ativan | Loravan | Milzepam | Slivan;
- (KW) Kuwait: Apo lorazepam | Ativan;
- (LB) Lebanon: Ativan | Lorapam;
- (LT) Lithuania: Lorafen | Lorans | Lorazepam polta | Lorazepam tzf;
- (LU) Luxembourg: Docloraze | Lauracalm | Lorazepam Eurogenerics | Lorazepam mylan | Lori-

- dem | Serenase | Temesta | Vigiten;
- (LV) Latvia: Lorafen;
 - (MA) Morocco: Lorans | Temesta;
 - (MX) Mexico: Apolazam | Ativan | Lozam | Plimbox | Sinestron;
 - (MY) Malaysia: Apo lorazepam | Ativan | Lorans | Tranpam;
 - (NL) Netherlands: Lorazepam A | Lorazepam Alharma | Lorazepam flx | Lorazepam prolepha | Temesta;
 - (NO) Norway: Lorazepam aa pharma | Lorazepam aristo | Lorazepam aurobindo | Lorazepam dorom | Lorazepam eg | Lorazepam leading | Lorazepam nuraxpharm | Lorazepam orifarm | Lorazepam orion | Rativor | Temesta;
 - (NZ) New Zealand: Ativan | Lorapam | Lorzem;
 - (PE) Peru: Ativan;
 - (PH) Philippines: Ativan;
 - (PK) Pakistan: Arivan | Arsopal | Ativan | Avor | Emotivan | Gtvan | L pam | Lipaz | Lo aram | Loram | Medi-s | Orgalopam | Revel | Tenzil | Tranquil;
 - (PL) Poland: Lorabex | Lorafen | Lorazepam orion | Lorazepam tzf | Temesta;
 - (PR) Puerto Rico: Ativan | Loreev xr;
 - (PT) Portugal: Ansilor | Lorazepam aristo | Lorazepam Pharmakern | Lorenin | Lersedal;
 - (QA) Qatar: Ativan | Lorans | Lorivan;
 - (RO) Romania: Anxiar | Lorivan;
 - (RU) Russian Federation: Lorafen | Merlit;
 - (SA) Saudi Arabia: Ativan;
 - (SE) Sweden: Lorazepam orifarm | Lorazepam orion | Temesta;
 - (SG) Singapore: Apo lorazepam | Ativan | Lorans;
 - (SI) Slovenia: Lorafen | Loram | Lorazepam Kern pharma | Lorsilan | Temesta;
 - (SK) Slovakia: Tavor expidet;
 - (SR) Suriname: Apo lorazepam | Larpose | Lorivan;
 - (SV) El Salvador: Xorax;
 - (TH) Thailand: Anta | Anxira | Ativan | Hawkcopax | Kemora | Lonza | Lora | Lora-p | Loramed | Lorans | Lorapam | Lorapine | Lorazene | Lorazep | Lorazin | Lozepine | Ora | Razepam | Tranavan | Vanmed;
 - (TN) Tunisia: Loram | Lorans | Temesta;
 - (TR) Turkey: Ativan;
 - (TW) Taiwan: Anxicam | Anxiedin | Anzepam | Aramin | Atilen | Atipam | Ativan | Atizepam | Control | Fuletin | Kolinin | Larbam | Lopam | Lorapam | Lorazepam | Lorazin | Lorpam | Lovamin | Lowen | Neuropam | Padelin | Quait | Silence | Spolin | Stapam | Wintin | Wypax;
 - (UY) Uruguay: Atemperator | Ativan | Farmazepam | Lorazepam lazar | Lorazepam | Lorazepam Chobet | Solepar | Tranplus | Trapax;
 - (VE) Venezuela, Bolivarian Republic of: Ativan;
 - (ZA) South Africa: Ativan | Rolab-lorazepam | Tranqipam;
 - (ZW) Zimbabwe: Apo lorazepam

Use of UpToDate is subject to the Terms of Use.

REFERENCES

1. 2023 American Geriatrics Society Beers Criteria Update Expert Panel. American Geriatrics Society 2023 updated AGS Beers Criteria for potentially inappropriate medication use in older adults. *J Am Geriatr Soc.* 2023;71(7):2052-2081. doi:10.1111/jgs.18372 [PubMed 37139824]
2. Abernethy DR, Greenblatt DJ, Divoll M, Shader RI. Enhanced glucuronide conjugation of drugs in obesity: studies of lorazepam, oxazepam, and acetaminophen. *J Lab Clin Med.* 1983;101(6):873-880. [PubMed 6133901]
3. Adler L, Angrist B, Peselow E, Corwin J, Rotrosen J. Efficacy of propranolol in neuroleptic-induced akathisia. *J Clin Psychopharmacol.* 1985;5(3):164-166. [PubMed 2860136]

4. Agulló-García A, Garcés Sotillos M, Colás Sanz C. Fixed drug eruption due to lorazepam. *J Invest Allergol Clin Immunol*. 2018;28(3):185-186. doi:10.18176/jiaci.0225 [PubMed 29939135]
5. Ahlfors CE. Benzyl alcohol, kernicterus, and unbound bilirubin. *J Pediatr*. 2001;139(2):317-319. [PubMed 11487763]
6. Ahmad S, Ellis JC, Kamwendo H, Molyneux E. Efficacy and safety of intranasal lorazepam versus intramuscular paraldehyde for protracted convulsions in children: an open randomised trial. *Lancet*. 2006;367(9522):1591-1597. [PubMed 16698412]
7. Allison C, Pratt JA. Neuroadaptive processes in GABAergic and glutamatergic systems in benzodiazepine dependence. *Pharmacol Ther*. 2003;98(2):171-195. doi:10.1016/s0163-7258(03)00029-9 [PubMed 12725868]
8. Akerele E, Olupona T. Drugs of abuse. *Psychiatr Clin North Am*. 2017;40(3):501-517. doi:10.1016/j.psc.2017.05.006 [PubMed 28800805]
9. American College of Obstetricians and Gynecologists (ACOG). Treatment and management of mental health conditions during pregnancy and postpartum: ACOG clinical practice guideline No. 5. *Obstet Gynecol*. 2023;141(6):1262-1288. doi:10.1097/AOG.0000000000005202 [PubMed 37486661]
10. American Society of Addiction Medicine (ASAM). The ASAM clinical practice guideline on alcohol withdrawal management. *J Addict Med*. 2020;14(3S)(suppl 1):1-72. doi:10.1097/ADM.0000000000000668 [PubMed 32511109]
11. Amini A, Heidari K, Asadollahi S, et al. Intravenous promethazine versus lorazepam for the treatment of peripheral vertigo in the emergency department: A double blind, randomized clinical trial of efficacy and safety. *J Vestib Res*. 2014;24(1):39-47. doi:10.3233/VES-130506 [PubMed 24594499]
12. Amirnovin R, Sanchez-Pinto LN, Okuhara C, et al. Implementation of a risk-stratified opioid and benzodiazepine weaning protocol in a pediatric cardiac ICU. *Pediatr Crit Care Med*. 2018;19(11):1024-1032. doi:10.1097/PCC.0000000000001719 [PubMed 30234674]
13. Anand KJ, Hall RW. Pharmacological therapy for analgesia and sedation in the newborn. *Arch Dis Child Fetal Neonatal Ed*. 2006;91(6):F448-F453. doi:10.1136/adc.2005.082263 [PubMed 17056842]
14. Anderson PO, Sauberan JB. Modeling drug passage into human milk. *Clin Pharmacol Ther*. 2016;100(1):42-52. [PubMed 27060684]
15. Andrade C, Varadharajan N, Bascarane S, Menon V. Gestational exposure to benzodiazepines or z-hypnotics and neurodevelopmental disorders in offspring: systematic review and meta-analysis. *Acta Psychiatr Scand*. 2024;150(2):65-77. doi:10.1111/acps.13696 [PubMed 38751163]
16. Apo-Lorazepam [product monograph]. Toronto, Ontario, Canada: Apotex Inc; August 2013.
17. Arnold TC, Ryan ML. Acute amphetamine and synthetic cathinone ("bath salt") intoxication. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 30, 2019.
18. Aronoff GR, Bennett WM, Berns JS, et al, eds. *Drug Prescribing in Renal Failure: Dosing Guidelines for Adults and Children*. 5th ed. American College of Physicians; 2007:126, 182.
19. Arroliga AC, Shehab N, McCarthy K, Gonzales JP. Relationship of continuous infusion lorazepam to serum propylene glycol concentration in critically ill adults. *Crit Care Med*. 2004;32(8):1709-1714. [PubMed 15286548]
20. Arya R, Gulati S, Kabra M, Sahu JK, Kalra V. Intranasal versus intravenous lorazepam for control of acute seizures in children: a randomized open-label study. *Epilepsia*. 2011;52(4):788-793. [PubMed 21275979]

21. Ativan (lorazepam) [product monograph]. Kirkland, Quebec, Canada: Pfizer Canada ULC; December 2022.
22. Ativan (lorazepam) injection [prescribing information]. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; January 2023.
23. Ativan (lorazepam) tablet [prescribing information]. Bridgewater, NJ: Bausch Health US LLC; July 2025.
24. Authier N, Balayssac D, Sautereau M, et al. Benzodiazepine dependence: focus on withdrawal syndrome. *Ann Pharm Fr.* 2009;67(6):408-413. doi:10.1016/j.pharma.2009.07.001 [PubMed 19900604]
25. Bahn EL, Holt KR. Procedural sedation and analgesia: a review and new concepts. *Emerg Med Clin North Am.* 2005;23(2):503-517. doi:10.1016/j.emc.2004.12.013 [PubMed 15829394]
26. Bandelow B, Allgulander C, Baldwin DS, et al. World Federation of Societies of Biological Psychiatry (WFSBP) guidelines for treatment of anxiety, obsessive-compulsive and posttraumatic stress disorders - version 3. Part I: anxiety disorders. *World J Biol Psychiatry.* 2023;24(2):79-117. doi:10.1080/15622975.2022.2086295 [PubMed 35900161]
27. Barnes BJ, Gerst C, Smith JR, Terrell AR, Mullins ME. Osmol gap as a surrogate marker for serum propylene glycol concentrations in patients receiving lorazepam for sedation. *Pharmacotherapy.* 2006;26(1):23-33. [PubMed 16422667]
28. Barr J, Fraser GL, Puntillo K, et al; American College of Critical Care Medicine. Clinical practice guidelines for the management of pain, agitation, and delirium in adult patients in the intensive care unit. *Crit Care Med.* 2013;41(1):263-306. doi:10.1097/CCM.0b013e3182783b72 [PubMed 23269131]
29. Bartels M, Heide K, Mann K, Schied HW. Treatment of akathisia with lorazepam. An open clinical trial. *Pharmacopsychiatry.* 1987;20(2):51-53. doi:10.1055/s-2007-1017074 [PubMed 2884681]
30. Basura GJ, Adams ME, Monfared A, et al. Clinical practice guideline: Ménière's disease. *Otolaryngol Head Neck Surg.* 2020;162(2)(suppl):S1-S55. doi:10.1177/0194599820909438 [PubMed 32267799]
31. Becker DE. Pharmacodynamic considerations for moderate and deep sedation. *Anesth Prog.* 2012;59(1):28-42. doi:10.2344/0003-3006-59.1.28 [PubMed 22428972]
32. Bellantuono C, Tofani S, Di Sciascio G, Santone G. Benzodiazepine exposure in pregnancy and risk of major malformations: a critical overview. *Gen Hosp Psychiatry.* 2013;35(1):3-8. doi:10.1016/j.genhosppsy.2012.09.003 [PubMed 23044244]
33. Bhattacharyya N, Gubbels SP, Schwartz SR, et al. Clinical practice guideline: benign paroxysmal positional vertigo (update). *Otolaryngol Head Neck Surg.* 2017;156(3)(suppl):S1-S47. doi:10.1177/0194599816689667 [PubMed 28248609]
34. Blin O, Simon N, Jouve E, et al. Pharmacokinetic and pharmacodynamic analysis of sedative and amnesic effects of lorazepam in healthy volunteers. *Clin Neuropharmacol.* 2001;24(2):71-81. doi:10.1097/00002826-200103000-00002 [PubMed 11307041]
35. Bond A, Lader M. Differential effects of oxazepam and lorazepam on aggressive responding. *Psychopharmacology (Berl).* 1988;95(3):369-373. doi:10.1007/BF00181949 [PubMed 3137624]
36. Boyer EW. Serotonin syndrome (serotonin toxicity). Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 5, 2020.
37. Boyer EW. Methamphetamine: Acute intoxication. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 30, 2019b.
38. Brophy GM, Bell R, Claassen J, et al; Neurocritical Care Society Status Epilepticus Guideline Writing Committee. Guidelines for the evaluation and management of status epilepticus. *Neurocrit Care.* 2012;17(1):3-23. doi:10.1007/s12028-012-9695-z [PubMed 22528274]

39. Brunner E, Chen CA, Klein T, et al. Joint clinical practice guideline on benzodiazepine tapering: considerations when risks outweigh benefits. *J Gen Intern Med*. 2025;40(12):2814-2859. doi:10.1007/s11606-025-09499-2 [PubMed 40526204]
40. Bulathsinghala M, Keefer K, Van de Louw A. Trimethoprim/Sulfamethoxazole-induced severe lactic acidosis: A case report and review of the literature. *Medicine (Baltimore)*. 2016;95(17):e3478. doi:10.1097/MD.0000000000003478 [PubMed 27124045]
41. Burtles R, Astley B. Lorazepam in children. A double-blind trial comparing lorazepam, diazepam, trimeprazine and placebo. *Br J Anaesth*. 1983;55(4):275-279. [PubMed 6132612]
42. Bush G, Fink M, Petrides G, Dowling F, Francis A. Catatonia. II. Treatment with lorazepam and electroconvulsive therapy. *Acta Psychiatr Scand*. 1996;93(2):137-143. [PubMed 8686484]
43. Busto U, Sellers EM, Naranjo CA, Cappell H, Sanchez-Craig M, Sykora K. Withdrawal reaction after long-term therapeutic use of benzodiazepines. *N Engl J Med*. 1986;315(14):854-859. doi:10.1056/NEJM198610023151403 [PubMed 3092053]
44. Caillé G, Spénard J, Lacasse Y, Brennan J. Pharmacokinetics of two lorazepam formulations, oral and sublingual, after multiple doses. *Biopharm Drug Dispos*. 1983;4(1):31-42. doi:10.1002/bdd.2510040106 [PubMed 6132630]
45. Cammarano WB, Pittet JF, Weitz S, Schlobohm RM, Marks JD. Acute withdrawal syndrome related to the administration of analgesic and sedative medications in adult intensive care unit patients. *Crit Care Med*. 1998;26(4):676-84. doi:10.1097/00003246-199804000-00015 [PubMed 9559604]
46. Carson SS, Kress JP, Rodgers JE, et al. A randomized trial of intermittent lorazepam versus propofol with daily interruption in mechanically ventilated patients. *Crit Care Med*. 2006;34(5):1326-1332. doi: 10.1097/01.CCM.0000215513.63207.7F. [PubMed 16540958]
47. Centers for Disease Control (CDC). Neonatal deaths associated with use of benzyl alcohol—United States. *MMWR Morb Mortal Wkly Rep*. 1982;31(22):290-291. <http://www.cdc.gov/mmwr/preview/mmwr> [PubMed 6810084]
48. Cernaianu AC, DelRossi AJ, Flum DR, et al. Lorazepam and midazolam in the intensive care unit: a randomized, prospective, multicenter study of hemodynamics, oxygen transport, efficacy, and cost. *Crit Care Med*. 1996;24(2):222-228. doi:10.1097/00003246-199602000-00007 [PubMed 8605792]
49. Chamberlain JM, Capparelli EV, Brown KM, et al. Pharmacokinetics of intravenous lorazepam in pediatric patients with and without status epilepticus. *J Pediatr*. 2012;160(4):667-672. [PubMed 22050870]
50. Chang DH, Hiss S, Herich L, et al. Implantation of venous access devices under local anesthesia: patients' satisfaction with oral lorazepam. *Patient Prefer Adherence*. 2015;9:943-949. doi:10.2147/PPA.S80330 [PubMed 26185424]
51. Chicella M, Jansen P, Parthiban A, et al. Propylene glycol accumulation associated with continuous infusion of lorazepam in pediatric intensive care patients. *Crit Care Med*. 2002;30(12):2752-2756. doi:10.1097/00003246-200212000-00021 [PubMed 12483068]
52. Chouinard G. Issues in the clinical use of benzodiazepines: potency, withdrawal, and rebound. *J Clin Psychiatry*. 2004;65(suppl 5):7-12. [PubMed 15078112]
53. Choy Y. Acute procedural anxiety and specific phobia of clinical procedures in adults: Treatment overview. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 30, 2019.
54. Chuang HM, Meng LC, Lin CW, et al. Concomitant use of antidepressants and benzodiazepines during pregnancy and associated risk of congenital malformations: a population-based cohort study in Taiwan. *Lancet Psychiatry*. 2024;11(8):601-610. doi:10.1016/S2215-0366(24)00176-7 [PubMed 38968942]

55. Coffey MJ. Catatonia in adults: Epidemiology, clinical features, assessment, and diagnosis. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed October 23, 2023a.
56. Coffey MJ. Catatonia: Treatment and prognosis. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed October 23, 2023b.
57. Cosci F, Chouinard G. Acute and persistent withdrawal syndromes following discontinuation of psychotropic medications. *Psychother Psychosom*. 2020;89(5):283-306. doi:10.1159/000506868 [PubMed 32259826]
58. Craske M, Bystritsky A. Generalized anxiety disorder in adults: management. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed January 19, 2022.
59. Crawford TO, Mitchell WG, Snodgrass SR. Lorazepam in Childhood Status Epilepticus and Serial Seizures: Effectiveness and Tachyphylaxis. *Neurology*. 1987;37(2):190-195. [PubMed 3808299]
60. Curley MA, Wypij D, Watson RS, et al. Protocolized sedation vs usual care in pediatric patients mechanically ventilated for acute respiratory failure: a randomized clinical trial. *JAMA*. 2015;313(4):379-389. doi:10.1001/jama.2014.18399 [PubMed 25602358]
61. Curran HV. Benzodiazepines, memory and mood: a review. *Psychopharmacology (Berl)*. 1991;105(1):1-8. doi:10.1007/BF02316856 [PubMed 1684055]
62. Curran HV, Schiwy W, Lader M. Differential amnesic properties of benzodiazepines: a dose-response comparison of two drugs with similar elimination half-lives. *Psychopharmacology (Berl)*. 1987;92(3):358-364. doi:10.1007/BF00210844 [PubMed 3114788]
63. Daniels J. Catatonia: clinical aspects and neurobiological correlates. *J Neuropsychiatry Clin Neurosci*. 2009;21(4):371-380. doi:10.1176/appi.neuropsych.21.4.371. [PubMed 19996245]
64. Dawson J, Boyle J, Stanley N, Johnsen S, Hindmarch I, Skene DJ. Benzodiazepine-induced reduction in activity mirrors decrements in cognitive and psychomotor performance. *Hum Psychopharmacol*. 2008;23(7):605-613. doi:10.1002/hup.961 [PubMed 18570226]
65. Delcò F, Tchambaz L, Schlienger R, Drewe J, Krähenbühl S. Dose adjustment in patients with liver disease. *Drug Saf*. 2005;28(6):529-545. doi:10.2165/00002018-200528060-00005 [PubMed 15924505]
66. Deshmukh A, Wittert W, Schnitzler E, Mangurten HH. Lorazepam in the treatment of refractory neonatal seizures. A pilot study. *Am J Dis Child*. 1986;140(10):1042-1044. [PubMed 3752014]
67. Devlin JW, Skrobik Y, Gélinas C, et al. Clinical practice guidelines for the prevention and management of pain, agitation/sedation, delirium, immobility, and sleep disruption in adult patients in the ICU. *Crit Care Med*. 2018;46(9):e825-e873. doi:10.1097/CCM.0000000000003299 [PubMed 30113379]
68. Díaz-Gutiérrez MJ, Martínez-Cengotitabengoa M, Sáez de Adana E, et al. Relationship between the use of benzodiazepines and falls in older adults: A systematic review. *Maturitas*. 2017;101:17-22. doi:10.1016/j.maturitas.2017.04.002 [PubMed 28539164]
69. Dolder CR, Nelson MH. Hypnosedative-induced complex behaviours: incidence, mechanisms and management. *CNS Drugs*. 2008;22(12):1021-1036. doi:10.2165/0023210-200822120-00005. [PubMed 18998740]
70. Dominguez KD, Crowley MR, Coleman DM, Katz RW, Wilkins DG, Kelly HW. Withdrawal from lorazepam in critically ill children. *Ann Pharmacother*. 2006;40(6):1035-1039. doi:10.1345/aph.1G701 [PubMed 16720707]
71. Donaldson M, Gizzarelli G, Chanpong B. Oral sedation: a primer on anxiolysis for the adult patient. *Anesth Prog*. 2007;54(3):118-129. doi:10.2344/0003-3006(2007)54[118:OSAPOA]2.0.CO;2 [PubMed 17900211]
72. Drappatz J, Avila EK. Seizures in patients with primary and metastatic brain tumors. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 15, 2019.

73. Drislane FW. Convulsive status epilepticus in adults: management. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 29, 2025.
74. Dupuis LL, Robinson PD, Boodhan S, et al; Pediatric Oncology Group of Ontario. Guideline for the prevention and treatment of anticipatory nausea and vomiting due to chemotherapy in pediatric cancer patients. *Pediatr Blood Cancer*. 2014;61(8):1506-1512. doi: 10.1002/pbc.25063. [PubMed 24753095]
75. Dvořáčková E, Pilková A, Matoulek M, Slanař O, Hartinger JM. Bioavailability of orally administered drugs after bariatric surgery. *Curr Obes Rep*. 2024;13(1):141-153. doi:10.1007/s13679-023-00548-7 [PubMed 38172482]
76. Eleftheriou G, Zandonella Callegher R, et al. Consensus panel recommendations for the pharmacological management of breastfeeding women with postpartum depression. *Int J Environ Res Public Health*. 2024;21(5):551. doi:10.3390/ijerph21050551 [PubMed 38791766]
77. England ML, Ongür D, Konopaske GT, Karmacharya R. Catatonia in psychotic patients: clinical features and treatment response. *J Neuropsychiatry Clin Neurosci*. 2011;23(2):223-226. [PubMed 21677256]
78. Expert opinion. Senior Hepatic Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.
79. Expert opinion. Senior Obesity Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.
80. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.
81. Fink M, Taylor MA. The catatonia syndrome: forgotten but not gone. *Arch Gen Psychiatry*. 2009;66(11):1173-7. doi:10.1001/archgenpsychiatry.2009.141 [PubMed 19884605]
82. Fishman DS, Andropoulos DB, Lightdale JR. Sedation and the Food and Drug Administration Warning: What a Pediatric Gastroenterologist, Hepatologist, and Pancreatologist Should Know. *J Pediatr Gastroenterol Nutr*. 2019;69(1):3-5. doi:10.1097/MPG.0000000000002346 [PubMed 30921252]
83. Flint AJ. Generalised anxiety disorder in elderly patients: epidemiology, diagnosis and treatment options. *Drugs Aging*. 2005;22(2):101-114. doi:10.2165/00002512-200522020-00002 [PubMed 15733018]
84. Food and Drug Administration. General Anesthetic and sedation drugs; drug safety communication - FDA approves label changes for use in young children. Food and Drug Administration website. Accessed July 6, 2017. <https://www.fda.gov/drugs/drug-safety-and-availability/fda-drug-safety-communication-fda-approves-label-changes-use-general-anesthetic-and-sedation-drugs>.
85. Food and Drug Administration. FDA Drug Safety Communication: FDA review results in new warnings about using general anesthetics and sedation drugs in young children and pregnant women. Food and Drug Administration website. Published December 14, 2016. Updated March 8, 2018. <https://www.fda.gov/drugs/drug-safety-and-availability/fda-drug-safety-communication-fda-review-results-new-warnings-about-using-general-anesthetics-and>
86. Franck LS, Scoppettuolo LA, Wypij D, Curley MAQ. Validity and generalizability of the Withdrawal Assessment Tool-1 (WAT-1) for monitoring iatrogenic withdrawal syndrome in pediatric patients. *Pain*. 2012;153(1):142-148. doi:10.1016/j.pain.2011.10.003 [PubMed 22093817]
87. Frank C. Recognition and treatment of serotonin syndrome. *Can Fam Physician*. 2008;54(7):988-992. [PubMed 18625822]
88. Fraser AA, Ingram IM. Lorazepam dependence and chronic psychosis. *Br J Psychiatry*. 1985;147:211. [PubMed 4041699]

89. Freeman MP, Góez-Mogollón L, McInerney KA, et al. Obstetrical and neonatal outcomes after benzodiazepine exposure during pregnancy: results from a prospective registry of women with psychiatric disorders. *Gen Hosp Psychiatry*. 2018;53:73-79. doi:10.1016/j.genhosppsych.2018.05.010 [PubMed 29958100]
90. Fuchs B, Bellamy C. Sedative-analgesia in ventilated adults: Medication properties, dose regimens, and adverse effects. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed January 11, 2022.
91. Furman JM, Barton JJS. Treatment of vertigo. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed November 28, 2022.
92. Gershkovich P, Wasan KM, Ribeyre C, Ibrahim F, McNeill JH. Effect of variations in treatment regimen and liver cirrhosis on exposure to benzodiazepines during treatment of alcohol withdrawal syndrome. *Drugs Context*. 2015;4:212287. doi:10.7573/dic.212287 [PubMed 26322116]
93. Gerson R, Malas N, Feuer V, Silver GH, Prasad R, Mroczkowski MM. Best practices for evaluation and treatment of agitated children and adolescents (BETA) in the emergency department: consensus statement of the American Association for Emergency Psychiatry. *West J Emerg Med*. 2019;20(2):409-418. [PubMed 30881565]
94. Giersch A, Boucart M, Elliott M, Vidailhet P. Atypical behavioural effects of lorazepam: clues to the design of novel therapies? *Pharmacol Ther*. 2010;126(1):94-108. doi:10.1016/j.pharmthera.2010.01.00 [PubMed 20138190]
95. Glauser T, Shinnar S, Gloss D, et al. Evidence-based guideline: Treatment of convulsive status epilepticus in children and adults: report of the Guideline Committee of the American Epilepsy Society. *Epilepsy Curr*. 2016;16(1):48-61. doi:10.5698/1535-7597-16.1.48 [PubMed 26900382]
96. Goldney RD. Paradoxical reaction to a new minor tranquilizer. *Med J Aust*. 1977;1(5):139-40. [PubMed 15198]
97. Grabowski J, Goldin A, Arthur LG, et al. The effects of early anesthesia on neurodevelopment: A systematic review. *J Pediatr Surg*. 2021:S0022-3468(21)00014-2. doi:10.1016/j.jpedsurg.2021.01.002 [PubMed 33509654]
98. Graham MR. Clinical update regarding general anesthesia-associated neurotoxicity in infants and children. *Curr Opin Anaesthesiol*. 2017;30(6):682-687. doi:10.1097/ACO.0000000000000520 [PubMed 28915132]
99. Graves L, Carson G, Poole N, et al. Guideline no. 405: screening and counselling for alcohol consumption during pregnancy. *J Obstet Gynaecol Can*. 2020;42(9):1158-1173.e1. doi:10.1016/j.jogc.2020.03.002 [PubMed 32900457]
100. Graves NM, Kriel RL, Jones-Saete C. Bioavailability of rectally administered lorazepam. *Clin Neuropharmacol*. 1987;10(6):555-559. doi:10.1097/00002826-198712000-00007 [PubMed 3427562]
101. Greenblatt DJ, Divoll M, Harmatz JS, Shader RI. Pharmacokinetic comparison of sublingual lorazepam with intravenous, intramuscular, and oral lorazepam. *J Pharm Sci*. 1982;71(2):248-252. doi:10.1002/jps.2600710227 [PubMed 6121043]
102. Greenblatt DJ, Shader RI, and Abernethy DR. Drug therapy. Current status of benzodiazepines. *N Engl J Med*. 1983;309(6):354-358. [PubMed 6135156]
103. Griffin CE 3rd, Kaye AM, Bueno FR, Kaye AD. Benzodiazepine pharmacology and central nervous system-mediated effects. *Ochsner J*. 2013;13(2):214-223. [PubMed 23789008]
104. Grigoriadis S, Graves L, Peer M, et al. Benzodiazepine use during pregnancy alone or in combination with an antidepressant and congenital malformations: systematic review and meta-analysis. *J Clin Psychiatry*. 2019;80(4):18r12412. doi:10.4088/JCP.18r12412 [PubMed 31294935]
105. Grønbaek L, Watson H, Vilstrup H, Jepsen P. Benzodiazepines and risk for hepatic encephalopathy in patients with cirrhosis and ascites. *United European Gastroenterol J*. 2018;6(3):407-412.

doi:10.1177/2050640617727179 [PubMed 29774154]

106. Guerra GG, Robertson CM, Alton GY, et al; Western Canadian Complex Pediatric Therapies Follow-up Group. Neurodevelopmental outcome following exposure to sedative and analgesic drugs for complex cardiac surgery in infancy. *Paediatr Anaesth*. 2011;21(9):932-941. doi:10.1111/j.1460-9592.2011.03581.x [PubMed 21507125]
107. Hain TC, Uddin M. Pharmacological treatment of vertigo. *CNS Drugs*. 2003;17(2):85-100. doi:10.2165/00023210-200317020-00002. [PubMed 12521357]
108. Hanna J, Swetter S. A case of delirium and rhabdomyolysis in severe Iatrogenic opioid withdrawal. *Psychosomatics*. 2018;59(4):405-407. doi:10.1016/j.psym.2017.12.002 [PubMed 29325983]
109. Hansen L, Lange R, Gupta S. Development and evaluation of a guideline for monitoring propylene glycol toxicity in pediatric intensive care unit patients receiving continuous infusion lorazepam. *J Pediatr Pharmacol Ther*. 2015;20(5):367-372. doi:10.5863/1551-6776-20.5.367 [PubMed 26472950]
110. Harman SM, Walling AM. Palliative care: the last hours and days of life. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed August 14, 2025.
111. Hasan A, Falkai P, Wobrock T, et al; World Federation of Societies of Biological Psychiatry (WFSBP) Task Force on Treatment Guidelines for Schizophrenia. World Federation of Societies of Biological Psychiatry (WFSBP) guidelines for biological treatment of schizophrenia, part 1: update 2012 on the acute treatment of schizophrenia and the management of treatment resistance. *World J Biol Psychiatry*. 2012;13(5):318-378. doi:10.3109/15622975.2012.696143. [PubMed 22834451]
112. Hasan A, Falkai P, Wobrock T, et al; WFSBP Task Force on Treatment Guidelines for Schizophrenia. World Federation of Societies of Biological Psychiatry (WFSBP) guidelines for biological treatment of schizophrenia, part 2: update 2012 on the long-term treatment of schizophrenia and management of antipsychotic-induced side effects. *World J Biol Psychiatry*. 2013;14(1):2-44. doi:10.3109/15622975.2012.739708. [PubMed 23216388]
113. Hegenbarth MA; American Academy of Pediatrics Committee on Drugs. Preparing for pediatric emergencies: drugs to consider. *Pediatrics*. 2008;121(2):433-443. [PubMed 18245435]
114. Henry DW, Burwinkle JW, Klutman NE. Determination of sedative and amnestic doses of lorazepam in children. *Clin Pharm*. 1991;10(8):625-629. [PubMed 1934919]
115. Hesketh PJ, Kris MG, Basch E, et al. Antiemetics: ASCO guideline update. *J Clin Oncol*. 2020;38(24):2782-2797. doi:10.1200/JCO.20.01296 [PubMed 32658626]
116. Hilt RJ, Woodward TA. Agitation treatment for pediatric emergency patients. *J Am Acad Child Adolesc Psychiatry*. 2008;47(2):132-138. [PubMed 18216715]
117. Hoffman RS, Weinhouse GL. Management of moderate and severe alcohol withdrawal syndromes. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed March 24, 2026.
118. Holt S, Tetrault J. Alcohol withdrawal: ambulatory management. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed July 28, 2023.
119. Horinek EL, Kiser TH, Fish DN, MacLaren R. Propylene glycol accumulation in critically ill patients receiving continuous intravenous lorazepam infusions. *Ann Pharmacother*. 2009;43(12):1964-1971. doi:10.1345/aph.1M313 [PubMed 19920159]
120. Horn E, Nesbit SA. Pharmacology and pharmacokinetics of sedatives and analgesics. *Gastrointest Endosc Clin N Am*. 2004;14(2):247-268. doi:10.1016/j.giec.2004.01.001 [PubMed 15121142]
121. Howard P, Twycross R, Shuster J, Mihalyo M, Wilcock A. Benzodiazepines. *J Pain Symptom Manage*. 2014;47(5):955-964. doi: 10.1016/j.jpainsymman.2014.03.001. [PubMed 24681184]

122. Ikonomidou C, Bosch F, Miksa M, et al. Blockade of NMDA receptors and apoptotic neurodegeneration in the developing brain. *Science*. 1999;283(5398):70-4. doi:10.1126/science.283.5398.70 [PubMed 9872743]
123. "Inactive" ingredients in pharmaceutical products: update (subject review). American Academy of Pediatrics (AAP) Committee on Drugs. *Pediatrics*. 1997;99(2):268-278. [PubMed 9024461]
124. Iqbal MM, Sobhan T, Ryals T, et al. Effects of Commonly Used Benzodiazepines on the Fetus, the Neonate, and the Nursing Infant. *Psychiatr Serv*. 2002;53(1):39-49. [PubMed 11773648]
125. Iqbal O'Meara AM, Miller Ferguson N, Zven SE, et al. Potential neurodevelopmental effects of pediatric intensive care sedation and analgesia: Repetitive benzodiazepine and opioid exposure alters expression of glial and synaptic proteins in juvenile rats. *Crit Care Explor*. 2020;2(4):e0105. doi:10.1097/CCE.000000000000105 [PubMed 32426747]
126. Ito S. Drug therapy for breast-feeding women. *N Engl J Med*. 2000;343(2):118-126. [PubMed 10891521]
127. Jahn A, Bodreau C, Farthing K, Elbarbry F. Assessing propylene glycol toxicity in alcohol withdrawal patients receiving intravenous benzodiazepines: A one-compartment pharmacokinetic model. *Eur J Drug Metab Pharmacokinet*. 2018;43(4):423-430. doi: 10.1007/s13318-018-0462-1 [PubMed 29392569]
128. Jennum P, Baandrup L, Ibsen R, et al. Increased all-cause mortality with use of psychotropic medication in dementia patients and controls: A population-based register study. *Eur Neuropsychopharmacol*. 2015;25(11):1906-1913. doi:10.1016/j.euroneuro.2015.08.014 [PubMed 26342397]
129. Jirsch J, Hirsch LJ. Nonconvulsive status epilepticus: treatment and prognosis. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed April 25, 2025.
130. Johnson TJ, Voss G. Continuous infusion of undiluted lorazepam injection. *Am J Health Syst Pharm*. 2002;59(1):78-79. [PubMed 11813472]
131. Kalachnik JE, Hanzel TE, Sevenich R, Harder SR. Benzodiazepine behavioral side effects: review and implications for individuals with mental retardation. *Am J Ment Retard*. 2002;107(5):376-410. doi:10.1352/0895-8017(2002)107<0376:BBSERA>2.0.CO;2 [PubMed 12186578]
132. Kanto JH. Use of benzodiazepines during pregnancy, labour and lactation, with particular reference to pharmacokinetic considerations. *Drugs*. 1982;23(5):354-380. doi:10.2165/00003495-198223050-00002 [PubMed 6124415]
133. Kapur J, Elm J, Chamberlain JM, et al. Randomized trial of three anticonvulsant medications for status epilepticus. *N Engl J Med*. 2019;381(22):2103-2113. doi:10.1056/NEJMoa1905795 [PubMed 31774955]
134. Katzman MA, Bleau P, Blier P, Chokka P, Kjernisted K, Van Ameringen M; Canadian Anxiety Guidelines Initiative Group. Canadian clinical practice guidelines for the management of anxiety, post-traumatic stress and obsessive-compulsive disorders. *BMC Psychiatry*. 2014;14(suppl 1):S1. doi:10.1186/1471-244X-14-S1-S1 [PubMed 25081580]
135. Kelly LE, Poon S, Madadi P, Koren G. Neonatal benzodiazepines exposure during breastfeeding. *J Pediatr*. 2012;161(3):448-451. [PubMed 22504099]
136. Khaldarov N. Benzodiazepines for treatment of neuroleptic malignant syndrome. *Hosp Physician*. 2000;36(9):51-55.
137. Kirkby KC, Montgomery IM, Badcock R, Daniels BA. A comparison of age-related deficits in memory and frontal lobe function following oral lorazepam administration. *J Psychopharmacol*. 1995;9(4):319-25. doi:10.1177/026988119500900405 [PubMed 22298396]
138. Kliegman RM, Stanton BF, St. Gemell JW, et al, eds. *Nelson Textbook of Pediatrics*. 19th ed. Saunders Elsevier; 2011.

139. Kliegman RM and St. Geme J, eds. *Nelson Textbook of Pediatrics*. 21st ed. Saunders Elsevier; 2020.
140. Kraus JW, Desmond PV, Marshall JP, Johnson RF, Schenker S, Wilkinson GR. Effects of aging and liver disease on disposition of lorazepam. *Clin Pharmacol Ther*. 1978;24(4):411-419. doi:10.1002/cpt1978244411 [PubMed 28871]
141. Lader M. Benzodiazepines revisited--will we ever learn? *Addiction*. 2011;106(12):2086-2109. doi:10.1111/j.1360-0443.2011.03563.x [PubMed 21714826]
142. Lavonas EJ. First-generation (typical) antipsychotic medication poisoning. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 21, 2021.
143. Lee WM, Lugo RA, Rusho WJ, Mackay M, Sweeley J. Chemical stability of extemporaneously prepared Lorazepam suspension at two temperatures. *J Pediatr Pharmacol Ther*. 2004;9(4):254-258.
144. Lemmer P, Schneider S, Mühe A, Wennig R. Quantification of lorazepam and lormetazepam in human breast milk using GC-MS in the negative chemical ionization mode. *J Anal Toxicol*. 2007;31(4):224-226. [PubMed 17555647]
145. Leppik IE, Patel SI. Intramuscular and rectal therapies of acute seizures. *Epilepsy Behav*. 2015;49:307-312. doi:10.1016/j.yebeh.2015.05.001 [PubMed 26071998]
146. Levin TT, Cortes-Ladino A, Weiss M, Palomba ML. Life-threatening serotonin toxicity due to a citalopram-fluconazole drug interaction: case reports and discussion. *Gen Hosp Psychiatry*. 2008;30(4):372-377. doi:10.1016/j.genhosppsy.2008.03.008 [PubMed 18585543]
147. Lim TY, Poole RL, Pageler NM. Propylene glycol toxicity in children. *J Pediatr Pharmacol Ther*. 2014;19(4):277-282. doi:10.5863/1551-6776-19.4.277 [PubMed 25762872]
148. Little JD, Taghavi EH. Disinhibition after lorazepam augmentation of antipsychotic medication. *Am J Psychiatry*. 1991;148(8):1099-100. doi:10.1176/ajp.148.8.1099a [PubMed 1677237]
149. Lohr L. Chemotherapy-induced nausea and vomiting. *Cancer J*. 2008;14(2):85-93. [PubMed 18391612]
150. Lonardo NW, Mone MC, Nirula R, et al. Propofol is associated with favorable outcomes compared with benzodiazepines in ventilated intensive care unit patients [published correction appears in *Am J Respir Crit Care Med*. 2014;189(11):e70]. *Am J Respir Crit Care Med*. 2014;189(11):1383-1394. doi:10.1164/rccm.201312-2291OC [PubMed 24720509]
151. Lorazepam injection [prescribing information]. Eatontown, NJ: West-ward Pharmaceuticals; June 2011.
152. Lorazepam injection USP solution 2 mg/mL (lorazepam) [product monograph]. Boucherville, Quebec, Canada: Sandoz Canada Inc; May 2021.
153. Lorazepam injection USP sterile solution 4 mg/mL (lorazepam) [product monograph]. Boucherville, Quebec, Canada: Sandoz Canada Inc; July 2023.
154. Lorazepam Intensol [prescribing information]. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; January 2023.
155. Lorazepam oral concentrate [prescribing information]. Berkeley Heights, NJ: Hikma Pharmaceuticals USA Inc; January 2023.
156. Lorazepam tablets [prescribing information]. Columbus, OH: American Health Packaging.
157. Loreev XR capsules [prescribing information]. Morristown, NJ: Almatica Pharma LLC; January 2023.
158. Lucca JM, Ramesh M, Parthasarathi G, Ram D. Lorazepam-induced diplopia. *Indian J Pharmacol*. 2014;46(2):228-229. doi:10.4103/0253-7613.129328 [PubMed 24741200]

159. Malcolm R, Myrick H, Roberts J, Wang W, Anton RF, Ballenger JC. The effects of carbamazepine and lorazepam on single versus multiple previous alcohol withdrawals in an outpatient randomized trial. *J Gen Intern Med.* 2002;17(5):349-355. [PubMed 12047731]
160. Male CG, Johnson HD. Oral benzodiazepine premedication in minor gynaecological surgery. *Br J Anaesth.* 1984;56(5):499-507. doi:10.1093/bja/56.5.499 [PubMed 6144317]
161. Mancuso CE, Tanzi MG, Gabay M. Paradoxical reactions to benzodiazepines: literature review and treatment options. *Pharmacotherapy.* 2004;24(9):1177-1185. [PubMed 15460178]
162. Marill KA, Walsh MJ, Nelson BK. Intravenous lorazepam versus dimenhydrinate for treatment of vertigo in the emergency department: a randomized clinical trial. *Ann Emerg Med.* 2000;36(4):310-319. doi:10.1067/mem.2000.110580 [PubMed 11020677]
163. Maytal J, Novak GP, King KC. Lorazepam in the treatment of refractory neonatal seizures. *J Child Neurol.* 1991;6(4):319-323. [PubMed 1940133]
164. McAllister-Williams RH, Baldwin DS, Cantwell R, et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. *J Psychopharmacol.* 2017;31(5):519-552. doi:10.1177/0269881117699361 [PubMed 28440103]
165. McCabe RE, Bui E. Specific phobia in adults: treatment overview. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed January 16, 2024.
166. McDermott CA, Kowalczyk AL, Schnitzler ER, et al. Pharmacokinetics of Lorazepam in Critically Ill Neonates With Seizures. *J Pediatr.* 1992;120(3):479-483. [PubMed 1538303]
167. McKee HR, Abou-Khalil B. Outpatient pharmacotherapy and modes of administration for acute repetitive and prolonged seizures. *CNS Drugs.* 2015;29(1):55-70. doi:10.1007/s40263-014-0219-6 [PubMed 25583219]
168. McPherson C. Sedation and analgesia in mechanically ventilated preterm neonates: continue standard of care or experiment? *J Pediatr Pharmacol Ther.* 2012;17(4):351-364. doi:10.5863/1551-6776-17.4.351 [PubMed 23413121]
169. Mejo SL. Anterograde amnesia linked to benzodiazepines. *Nurse Pract.* 1992;17(10):44, 49-50. doi:10.1097/00006205-199210000-00013 [PubMed 1357612]
170. Moore MJ, Im D. The acutely agitated or violent adult: pharmacologic management. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed December 4, 2024.
171. Moore TA. Ambulatory detoxification in alcohol use disorder and opioid use disorder. *Ment Health Clin.* 2020b;10(6):307-316. doi:10.9740/mhc.2020.11.307 [PubMed 33224688]
172. Morrison G, Chiang ST, Koepke HH, Walker BR. Effect of renal impairment and hemodialysis on lorazepam kinetics. *Clin Pharmacol Ther.* 1984;35(5):646-652. doi:10.1038/clpt.1984.89 [PubMed 6713774]
173. Mundeeler P. Child premedication per os with lorazepam. *Acta Anaesthesiol Belg.* 1980;(31)(suppl):187-193. [PubMed 6109421]
174. Myrick H, Malcolm R, Randall PK, et al. A double-blind trial of gabapentin versus lorazepam in the treatment of alcohol withdrawal. *Alcohol Clin Exp Res.* 2009;33(9):1582-1588. doi:10.1111/j.1530-0277.2009.00986.x [PubMed 19485969]
175. National Institute for Health and Care Excellence (NICE). Epilepsies: diagnosis and management. <https://www.nice.org.uk/guidance/cg137/resources/epilepsies-diagnosis-and-management-35109515407813> Published January 2012.
176. Neale BW, Mesler EL, Young M, Rebuck JA, Weise WJ. Propylene glycol-induced lactic acidosis in a patient with normal renal function: a proposed mechanism and monitoring recommendations. *Ann Pharmacother.* 2005;39(10):1732-1736. doi:10.1345/aph.1G083 [PubMed 16159998]

177. Nelsen JL, Haas CE, Habtemariam B, et al. A prospective evaluation of propylene glycol clearance and accumulation during continuous-infusion lorazepam in critically ill patients. *J Intensive Care Med*. 2008;23(3):184-194. doi:10.1177/0885066608315808 [PubMed 18543419]
178. Nelson J, Chouinard G; Canadian Society for Clinical Pharmacology. Guidelines for the clinical use of benzodiazepines: pharmacokinetics, dependency, rebound and withdrawal. *Can J Clin Pharmacol*. 1999;6(2):69-83. [PubMed 10519733]
179. Nelson L. Cocaine: Acute intoxication. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed January 24, 2022.
180. Nelson MA, Bradshaw WT. Use of benzodiazepines as anxiolytics in neonates: are we there yet?. *Adv Neonatal Care*. 2014;14(2):78-87. doi:10.1097/ANC.0000000000000049 [PubMed 24675625]
181. Nishimura A, Furugen A, Umazume T, et al. Benzodiazepine concentrations in the breast milk and plasma of nursing mothers: estimation of relative infant dose. *Breastfeed Med*. 2021;16(5):424-431. doi:10.1089/bfm.2020.0259 [PubMed 33449825]
182. Nutt DJ. Pharmacological mechanisms of benzodiazepine withdrawal. *J Psychiatr Res*. 1990;24 (Suppl 2):105-110. doi:10.1016/0022-3956(90)90041-n [PubMed 1980689]
183. O'Boyle CA. Benzodiazepine-induced amnesia and anaesthetic practice: a review. *Psychopharmacol Ser*. 1988;6:146-65. [PubMed 2905804]
184. O'Connor FG, Casa DJ. Exertional heat illness in adolescents and adults: Management and prevention. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed May 5, 2020.
185. O'Mahony D, Cherubini A, Guiteras AR, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 3. *Eur Geriatr Med*. 2023;14(4):625-632. doi:10.1007/s41999-023-00777-y [PubMed 37256475]
186. Owusu KA, Dhakar MB, Bautista C, et al. Comparison of intranasal midazolam versus intravenous lorazepam for seizure termination and prevention of seizure clusters in the adult epilepsy monitoring unit. *Epilepsy Behav*. 2019;98(pt A):161-167. doi:10.1016/j.yebeh.2019.07.021 [PubMed 31374472]
187. Palmaro A, Dupouy J, Lapeyre-Mestre M. Benzodiazepines and risk of death: results from two large cohort studies in France and UK. *Eur Neuropsychopharmacol*. 2015;25(10):1566-1577. doi:10.1016/j.euroneuro.2015.07.006 [PubMed 26256008]
188. Pandharipande PP, Pun BT, Herr DL, et al. Effect of sedation with dexmedetomidine vs lorazepam on acute brain dysfunction in mechanically ventilated patients: the MENDS randomized controlled trial. *JAMA*. 2007;298(22):2644-2653. [PubMed 18073360]
189. Pandit SK, Heisterkamp DV, Cohen PJ. Further studies of the anti-recall effect of lorazepam: a dose--time--effect relationship. *Anesthesiology*. 1976;45(5):495-500. [PubMed 9845]
190. Papini O, da Cunha SP, da Silva Mathes Ado C, et al. Kinetic disposition of lorazepam with focus on the glucuronidation capacity, transplacental transfer in parturients and racemization in biological samples. *J Pharm Biomed Anal*. 2006;40(2):397-403. doi:10.1016/j.jpba.2005.07.021 [PubMed 16143486]
191. Park TW. Benzodiazepine use disorder. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed November 25, 2025.
192. Patel P, Robinson PD, Devine KA, et al. Prevention and treatment of anticipatory chemotherapy-induced nausea and vomiting in pediatric cancer patients and hematopoietic stem cell recipients: clinical practice guideline update. *Pediatr Blood Cancer*. 2021;68(5):e28947. doi:10.1002/pbc.28947 [PubMed 33686754]
193. Peppers MP. Benzodiazepines for alcohol withdrawal in the elderly and in patients with liver disease. *Pharmacotherapy*. 1996;16(1):49-57. [PubMed 8700792]

194. Peters CG, Brunton JT. Comparative study of lorazepam and trimeprazine for oral premedication in paediatric anaesthesia. *Br J Anaesth*. 1982;54(6):623-628. [PubMed 6123337]
195. Pietras CJ, Lieving LM, Cherek DR, Lane SD, Tcheremissine OV, Nouvion S. Acute effects of lorazepam on laboratory measures of aggressive and escape responses of adult male parolees. *Behav Pharmacol*. 2005;16(4):243-51. doi:10.1097/01.fbp.0000170910.53415.77 [PubMed 15961964]
196. Rabinstein AA. Neuroleptic malignant syndrome. Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <https://www.uptodate.com>. Accessed May 23, 2025.
197. Radojčić MR, El Marroun H, Miljković B, et al. Prenatal exposure to anxiolytic and hypnotic medication in relation to behavioral problems in childhood: a population-based cohort study. *Neurotoxicol Teratol*. 2017;61:58-65. doi:10.1016/j.ntt.2017.02.005 [PubMed 28259732]
198. Rajiv KR, Radhakrishnan A. Status epilepticus in pregnancy - can we frame a uniform treatment protocol? *Epilepsy Behav*. 2019;101(Pt B):106376. doi:10.1016/j.yebeh.2019.06.020 [PubMed 31303443]
199. Reddy MSS, Achary U, Harbishettar V, Sivakumar PT, Varghese M. Paradoxical reaction to benzodiazepines in elderly - Case series. *Asian J Psychiatr*. 2018;35:8-10. doi:10.1016/j.ajp.2018.04.037 [PubMed 29723722]
200. Refer to manufacturer's labeling.
201. Richardson JK, Eckner JT, Kim H, Ashton-Miller JA. A clinical method of evaluating simple reaction time and reaction accuracy is sensitive to a single dose of lorazepam. *J Psychopharmacol*. 2020;34(8):920-925. doi:10.1177/0269881120915409 [PubMed 32536331]
202. Rogers JP, Oldham MA, Fricchione G, et al. Evidence-based consensus guidelines for the management of catatonia: recommendations from the British Association for Psychopharmacology. *J Psychopharmacol*. 2023;37(4):327-369. doi:10.1177/02698811231158232 [PubMed 37039129]
203. Rosebush PI, Mazurek MF. Catatonia and its treatment. *Schizophr Bull*. 2010;36(2):239-242. [PubMed 19969591]
204. Roth T, Roehrs T, Wittig R, Zorick F. Benzodiazepines and memory. *Br J Clin Pharmacol*. 1984;18 Suppl 1(Suppl 1):45S-49S. doi:10.1111/j.1365-2125.1984.tb02581.x [PubMed 6151849]
205. Roy-Byrne PP. Management of psychiatric disorders in patients with cancer. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed November 2, 2021.
206. Rozendaal FW, Spronk PE, Snellen FF, et al; UltiSAFE investigators. Remifentanyl-propofol analgo-sedation shortens duration of ventilation and length of ICU stay compared to a conventional regimen: a centre randomised, cross-over, open-label study in the Netherlands. *Intensive Care Med*. 2009;35(2):291-298. doi: 10.1007/s00134-008-1328-9. [PubMed 18949456]
207. Saarelainen L, Tolppanen AM, Koponen M, et al. Risk of death associated with new benzodiazepine use among persons with Alzheimer disease: A matched cohort study. *Int J Geriatr Psychiatry*. 2018;33(4):583-590. doi: 10.1002/gps.4821. [PubMed 29143367]
208. Sachdeva A, Chandra M, Deshpande SN. A comparative study of fixed tapering dose regimen versus symptom-triggered regimen of lorazepam for alcohol detoxification. *Alcohol Alcohol*. 2014;49(3):287-291. doi: 10.1093/alcalc/agt181. [PubMed 24407777]
209. Sansevere AJ, Bergin AM. Neonatal seizures. In: Eichenwald EC, Hansen AR, Martin CR, Stark AR. *Cloherty and Stark's Manual of Neonatal Care*. 9th ed. Lippincott Williams & Wilkins; 2022:chap. 56.
210. Schachter SC. Antiseizure medications: Mechanism of action, pharmacology, and adverse effects. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed December 8, 2020.

211. Scharf MB, Kales A, Bixler EO, Jacoby JA, Schweitzer PK. Lorazepam-efficacy, side effects, and rebound phenomena. *Clin Pharmacol Ther.* 1982;31(2):175-179. doi:10.1038/clpt.1982.27 [PubMed 6120058]
212. Shader RI, Dreyfuss D, Gerrein JR, Harmatz JS, Allison SJ, Greenblatt DJ. Sedative effects and impaired learning and recall after single oral doses of lorazepam. *Clin Pharmacol Ther.* 1986;39(5):526-529. doi:10.1038/clpt.1986.90 [PubMed 3698460]
213. Shehab N, Lewis CL, Streetman DD, Donn SM. Exposure to the pharmaceutical excipients benzyl alcohol and propylene glycol among critically ill neonates. *Pediatr Crit Care Med.* 2009;10(2):256-259. [PubMed 19188870]
214. Sheno RP, Timm N; Committee on Drugs; Committee on Pediatric Emergency Medicine. Drugs used to treat pediatric emergencies. *Pediatrics.* 2020;145(1):e20193450. [PubMed 31871244]
215. Shih G, Wallace R. Induced abortion in the first trimester: Procedures (uterine aspiration). Connor RF, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed December 8, 2020.
216. Short TG, Forrest P, Galletly DC. Paradoxical reactions to benzodiazepines--a genetically determined phenomenon? *Anaesth Intensive Care.* 1987;15(3):330-331. doi:10.1177/0310057X8701500314 [PubMed 3661967]
217. Society for Pediatric Sedation (SPS). FDA advisory on anesthesia and sedation medication in children. Society for Pediatric Sedation website. Accessed April 2021. [https://www.pedsedation.org/resources/safety/fda-advisory/#:~:text=In%20December%202016%2C%20the%20FDA,\(http%3A%2F%2Fwww.fda](https://www.pedsedation.org/resources/safety/fda-advisory/#:~:text=In%20December%202016%2C%20the%20FDA,(http%3A%2F%2Fwww.fda)
218. Solanki DI, Gohil JR, Patel AP. Comparative efficacy of phenobarbital, phenytoin and lorazepam for the treatment of neonatal seizures: a randomized trial. *J Clin Neonatol.* 2015;4:232-236.
219. Soyka M, Kranzler HR, Hesselbrock V, Kasper S, Mutschler J, Möller HJ; WFSBP Task Force on Treatment Guidelines for Substance Use Disorders. Guidelines for biological treatment of substance use and related disorders, part 1: Alcoholism, first revision. *World J Biol Psychiatry.* 2017;18(2):86-119. doi: 10.1080/15622975.2016.1246752. [PubMed 28006997]
220. Stein MB, Taylor CT. Social anxiety disorder in adults: Treatment overview. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed August 2, 2024.
221. Stewart SA. The effects of benzodiazepines on cognition. *J Clin Psychiatry.* 2005;66(suppl 2):9-13. [PubMed 15762814]
222. Stolbach A, Hoffman RS. Opioid withdrawal in the emergency setting. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed June 20, 2022.
223. Stroup TS. Schizophrenia in adults: Maintenance therapy and side effect management. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed October 5, 2021.
224. Summerfield RJ, Nielsen MS. Excretion of lorazepam into breast milk. *Br J Anaesth.* 1985;57(10):1042-1043. doi:10.1093/bja/57.10.1042-a [PubMed 4041315]
225. Sundbakk LM, Wood M, Gran JM, Nordeng H. Prenatal exposure to benzodiazepine and z-hypnotics and fifth-grade scholastic skills - emulating target trials using data from the Norwegian Mother, Father and Child Cohort Study. *Am J Epidemiol.* 2025;194(1):73-84. doi:10.1093/aje/kwae159 [PubMed 38944758]
226. Swart EL, de Jongh J, Zuideveld KP, Danhof M, Thijs LG, Strack van Schijndel RJ. Population pharmacokinetics of lorazepam and midazolam and their metabolites in intensive care patients on continuous venovenous hemofiltration. *Am J Kidney Dis.* 2005;45(2):360-371. doi:10.1053/j.ajkd.2004.09.004 [PubMed 15685515]
227. Swartz R, Longwell P. Treatment of vertigo. *Am Fam Physician.* 2005;71(6):1115-1122. [PubMed 15791890]

228. Szaflarski JP, Sadek A, Greve B, Williams P, Varner JA, Moseley BD. Randomized open-label trial of intravenous brivaracetam versus lorazepam for acute treatment of increased seizure activity. *Epilepsy Behav.* 2020;109:107127. doi:10.1016/j.yebeh.2020.107127 [PubMed 32417382]
229. Tan KR, Rudolph U, Lüscher C. Hooked on benzodiazepines: GABAA receptor subtypes and addiction. *Trends Neurosci.* 2011;34(4):188-197. doi:10.1016/j.tins.2011.01.004 [PubMed 21353710]
230. Tinker SC, Reefhuis J, Bitsko RH, et al; National Birth Defects Prevention Study. Use of benzodiazepine medications during pregnancy and potential risk for birth defects, National Birth Defects Prevention Study, 1997-2011. *Birth Defects Res.* 2019;111(10):613-620. doi:10.1002/bdr2.1497 [PubMed 30891943]
231. Tormoehlen LM, Rusyniak DE. Neuroleptic malignant syndrome and serotonin syndrome. *Handb Clin Neurol.* 2018;157:663-675. doi: 10.1016/B978-0-444-64074-1.00039-2. [PubMed 30459031]
232. Tsai MC, Huang TL. Severe neuroleptic malignant syndrome: successful treatment with high-dose lorazepam and diazepam: a case report. *Chang Gung Med J.* 2010;33(5):576-580. [PubMed 20979709]
233. Tsuda M, Shimizu N, Suzuki T. Contribution of glutamate receptors to benzodiazepine withdrawal signs. *Jpn J Pharmacol.* 1999;81(1):1-6. doi:10.1254/jjp.81.1 [PubMed 10580363]
234. Turkington D, Gill P. Mania induced by lorazepam withdrawal: a report of two cases. *J Affect Disord.* 1989;17(1):93-95. doi:10.1016/0165-0327(89)90028-1 [PubMed 2525581]
235. US Department of Veterans Affairs/Department of Defense (VA/DoD). VA/DoD clinical practice guideline for management of posttraumatic stress disorder and acute stress reaction. <https://www.healthquality.va.gov/HEALTHQUALITY/guidelines/mh/ptsd/index.asp>. Published 2023. Accessed November 27, 2023.
236. US Department of Veterans Affairs/US Department of Defense. VA/DoD clinical practice guideline for the management of substance use disorders. <https://www.limbic-cenc.org/wp-content/uploads/VA-DoD-CPG-Management-of-Substance-Use-Disorders-2021-Full-Report.pdf>. Published August 2021. Accessed May 12, 2022.
237. US Food and Drug Administration; Institute for Safe Medication Practices. FDA and ISMP Lists of Look-Alike Drug Names With Recommended Tall Man (Mixed Case) Letters. ISMP; 2023.
238. van Laar M, Volkerts E, Verbaten M. Subchronic effects of the GABA-agonist lorazepam and the 5-HT_{2A/2C} antagonist ritanserin on driving performance, slow wave sleep and daytime sleepiness in healthy volunteers. *Psychopharmacology (Berl).* 2001;154(2):189-197. doi:10.1007/s002130000633 [PubMed 11314681]
239. Van Sickle BJ, Xiang K, Tietz EI. Transient plasticity of hippocampal CA1 neuron glutamate receptors contributes to benzodiazepine withdrawal-anxiety. *Neuropsychopharmacology.* 2004;29(11):1994-2006. doi:10.1038/sj.npp.1300531 [PubMed 15266351]
240. Verbeeck R, Tjandramaga TB, Verberckmoes R, De Schepper PJ. Biotransformation and excretion of lorazepam in patients with chronic renal failure. *Br J Clin Pharmacol.* 1976;3(6):1033-1039. doi:10.1111/j.1365-2125.1976.tb00354.x [PubMed 22216526]
241. Vinkers CH, Olivier B. Mechanisms underlying tolerance after long-term benzodiazepine use: a future for subtype-selective gaba(a) receptor modulators? *Adv Pharmacol Sci.* 2012;2012:1-19. [PubMed 22536226]
242. von Gunten C, Buckholz G. Palliative care: Overview of cough, stridor, and hemoptysis. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. <http://www.uptodate.com>. Accessed April 29, 2019.
243. Wang RZ, Vashistha V, Kaur S, Houchens NW. Serotonin syndrome: Preventing, recognizing, and treating it. *Cleve Clin J Med.* 2016;83(11):810-817. doi:10.3949/ccjm.83a.15129 [PubMed 27824534]

244. Wang X, Zhang T, Ekheden I, et al. Prenatal exposure to benzodiazepines and Z-drugs in humans and risk of adverse neurodevelopmental outcomes in offspring: a systematic review. *Neurosci Biobehav Rev*. 2022;137:104647. doi:10.1016/j.neubiorev.2022.104647 [PubMed 35367514]
245. Wermeling DP, Miller JL, Archer SM, Manaligod JM, Rudy AC. Bioavailability and pharmacokinetics of lorazepam after intranasal, intravenous, and intramuscular administration. *J Clin Pharmacol*. 2001;41(11):1225-1231. doi:10.1177/00912700122012779 [PubMed 11697755]
246. Whitelaw AG, Cummings AJ, McFadyen IR. Effect of maternal lorazepam on the neonate. *Br Med J (Clin Res Ed)*. 1981;282(6270):1106-1108. [PubMed 6113019]
247. Wightman RS, Nelson LS, Lee JD, Fox LM, Smith SW. Severe opioid withdrawal precipitated by Vivitrol. *Am J Emerg Med*. 2018;36(6):1128.e1-1128.e2. doi:10.1016/j.ajem.2018.03.052 [PubMed 29605483]
248. Wikner BN, Stiller CO, Bergman U, et al. Use of benzodiazepines and benzodiazepine receptor agonists during pregnancy: neonatal outcome and congenital malformations. *Pharmacoepidemiol Drug Saf*. 2007;16(11):1203-1210. [PubMed 17894421]
249. Wilson KC, Reardon C, Farber HW. Propylene glycol toxicity in a patient receiving intravenous diazepam. *N Engl J Med*. 2000;343(11):815. doi:10.1056/nejm200009143431115 [PubMed 10991709]
250. Wilson S, Anderson K, Baldwin D, et al. British Association for Psychopharmacology consensus statement on evidence-based treatment of insomnia, parasomnias and circadian rhythm disorders: an update. *J Psychopharmacol*. 2019;33(8):923-947. doi:10.1177/0269881119855343 [PubMed 31271339]
251. Wodarz N, Krampe-Scheidler A, Christ M, et al. Evidence-based guidelines for the pharmacological management of acute methamphetamine-related disorders and toxicity. *Pharmacopsychiatry*. 2017;50(3):87-95. doi: 10.1055/s-0042-123752. [PubMed 28297728]
252. Wu HN, Liang Y, Li LL, Jiang HY, Xu LL. The safety of benzodiazepines and related drugs during pregnancy: an updated meta-analysis of cohort studies. *Arch Gynecol Obstet*. 2024;310(1):45-54. doi:10.1007/s00404-024-07557-4 [PubMed 38806942]
253. Yahwak JA, Riker RR, Fraser GL, Subak-Sharpe S. Determination of a lorazepam dose threshold for using the osmol gap to monitor for propylene glycol toxicity. *Pharmacotherapy*. 2008;28(8):984-991. [PubMed 18657015]
254. Yaucher NE, Fish JT, Smith HW, Wells JA. Propylene glycol-associated renal toxicity from lorazepam infusion. *Pharmacotherapy*. 2003;23(9):1094-1099. doi:10.1592/phco.23.10.1094.32762 [PubMed 14524641]
255. Zhong G, Wang Y, Zhang Y, Zhao Y. Association between benzodiazepine use and dementia: a meta-analysis. *PLoS One*. 2015;10(5):e0127836. doi:10.1371/journal.pone.0127836 [PubMed 26016483]
256. Ziegenbein M, Kropp S. Lorazepam-induced prolongation of the QT interval in a patient with schizoaffective disorder and complete AV block. *Can J Psychiatry*. 2004;49(6):414. doi:10.1177/070674370404900619 [PubMed 15283543]

Topic 9573 Version 958.0

Source: UpToDate. Downloaded 2026-05-26.